

Firm Boundaries and Produced Water Reuse in the Marcellus Shale

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Abstract

This paper presents evidence that trade in reusable wastewater among Pennsylvania oil and gas firms is subject to Coasean transaction costs. I first establish the empirical relevance of transaction costs by documenting changes in wastewater allocation within and across firms following a notable merger. I then build and estimate a transferable utility matching model in which firms substitute between internal reuse, trade with rivals, and conventional subsurface disposal. The estimates indicate that transaction costs are economically large but heterogeneous within and across relationships. To demonstrate the significance of these findings for environmental policy, I simulate the equilibrium effects of proposed changes to corrective taxes on subsurface disposal.

JEL: D22, D23, Q35, Q53, L23, L14, C71

Keywords: fracking; produced water; injection well disposal; transaction costs; matching

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1 Introduction

Oil and gas wells produce large volumes of wastewater. Like many environmentally consequential production decisions, wastewater management can be efficiently coordinated through markets. Yet market transactions are subject to Coasean transaction costs (Coase, 1937). This paper presents evidence that transaction costs prevent many otherwise efficient wastewater transfers, increasing firms’ reliance on controversial subsurface disposal wells. This finding has several implications for wastewater policy. Among other results, I show that failing to account for transaction costs can substantially bias estimates of key policy impacts.

I focus on Pennsylvania. Pennsylvania is now a major natural gas producer due to the adoption of hydraulic fracturing (“fracking”). Nearly all wastewater in the state is managed in one of two ways: through subsurface disposal or reuse in future fracking. To reuse wastewater, firms transport wastewater from geographically dispersed production wells to active fracking operations for which water is needed. Doing so efficiently requires that firms solve a matching problem: a given fracking operation requires only a finite volume of water, so reusing wastewater from one well reduces the capacity to reuse wastewater from another.

Wastewater “sharing”—trade for the purpose of reuse—is an important mechanism for optimizing match quality. Like all transactions between firms, wastewater sharing is potentially subject to frictions avoided under integration. These frictions can be described as transaction costs. To demonstrate the empirical relevance of transaction costs, I begin by reviewing the 2017 merger of two major natural gas producers, EQT and Rice. Prior to merging, the two firms never traded wastewater even though doing so appeared to offer large transportation cost savings. Afterwards, the merged firm began to transfer large volumes of reusable wastewater across the pre-merger firm boundary. This suggests that organizational frictions of some kind were the main barrier to pre-merger sharing, rather than transaction-specific production costs that would have persisted after the merger.

Motivated by this evidence, I next aim to quantify the overall empirical significance of Coasean transaction costs throughout the broader wastewater sharing market in Pennsylvania. In order to do so, I exploit detailed wastewater disposal records from the Pennsylvania Department of Environmental Protection (DEP). These records include all primary shipments of wastewater, including those that occur within the boundary of the firm. By a revealed preference-like argument similar to Atalay et al. (2019), this relatively rare combination of internal and external transaction data allows me to identify transaction costs in terms of the observable cost savings foregone when economic activity is kept inside the firm.

I obtain point estimates using a detailed empirical model of wastewater management in which reuse and sharing are represented by a standard model of one-to-one matching

with transferrable utility (Choo and Siow, 2006). In particular, I adapt the Choo and Siow (2006) model to the problem of resource allocation within and across firms by reducing each firm’s cost minimization problem to a collection of simultaneous, unit-by-unit discrete choice problems: firms select the least-cost destination for each new truckload of wastewater and, simultaneously, the least-cost source for each incremental truckload of water needed for ongoing fracking operations. Like in Shapley and Shubik (1971), the equilibrium match maximizes market-wide surplus: here, the sum of firms’ transportation, treatment, and transaction costs is minimized. The equilibrium match surplus is divided through endogenous transfer payments, while unmatched wastewater is transferred to subsurface disposal wells.

The model estimates clarify the magnitude of transaction costs across a wide range of potential sharing transactions. In a parsimonious specification of the model, I find that sharing one truckload of wastewater incurs transaction costs equivalent in magnitude to a 137.7 mile increase in shipment distance. In comparison, the average truckload of wastewater is shipped only 30.0 miles from its source. By this measure, transaction costs are economically large. Even among pairs of firms with positive sharing activity, transaction costs are equivalent in magnitude to a 68.9 mile increase in shipment distance. More flexible specifications reveal significant relationship-level heterogeneity in transaction costs which is poorly explained by observable firm characteristics such as firm size, location, and well geology.

This empirical approach not only allows me to quantify transaction costs, but also to explore their equilibrium implications. I focus on how transaction costs mediate firms’ reliance on subsurface disposal, a practice which has generated a range of environmental concerns in recent years (EPA, 2016; Groundwater Protection Council, 2019). In particular, I use the estimated model to simulate the effects of proposed disposal tax increases. The results suggest that a policymaker who did not account for transaction costs would overestimate impacts on disposal by an order of magnitude. This occurs because transaction costs are positively correlated with the observable costs of reuse, on account of longer shipment distances. Failure to account for transaction costs therefore understates the significance of unobserved sources of match surplus, making firms appear more sensitive to disposal cost changes.

The existence of transaction costs may or may not create inefficiency.¹ Nevertheless, measures designed to address transaction costs can potentially have large effects on generic wastewater-related inefficiencies, such as negative externalities from subsurface disposal. I conclude by exploring the value policies aimed at easing trade for a “portfolio” approach to environmental harms (Jaffe et al., 2005). Two areas of opportunity include greater public

¹As discussed in Section 8, the key issue for efficiency is whether transaction costs represent genuine economic costs or welfare-irrelevant decision costs (akin to the shadow costs of behavioral distortions). The distinction between welfare-relevant and welfare-irrelevant costs appears in the literature on switching costs in consumer markets (e.g., Handel and Kolstad, 2015). See also Buchanan and Stubblebine (1962).

disclosure of information about wastewater disposition and greater legal and regulatory certainty around wastewater ownership and liability (Groundwater Protection Council, 2019). Policies of this nature are unlikely to risk large rebound effects given the modest cost of wastewater management relative to other costs of extraction.

The remainder of the paper is structured as follows. This section continues with a discussion of contributions to the literature. Section 2 introduces the data and provides further institutional context. Section 3 presents the EQT-Rice merger analysis. Sections 4 and 5 present the model and estimation strategy, respectively. Section 6 presents the transaction cost estimates. Section 7 analyzes proposed disposal taxes. Section 8 discusses the broader significance of transaction costs for environmental policy. Section 9 concludes.

1.1 Related Literature

This paper bridges organizational and environmental economics by illustrating the empirical relevance of firm boundaries and Coasean transaction costs for environmentally sensitive production and disposal decisions in a particular industry. This contribution can be viewed in parallel to recent work in empirical industrial organization that builds on Buchanan (1969) to show how market power can materially affect various environmental outcomes (Preonas, 2023; Asker et al., 2024). Like in the case of market power, policy-relevant outcomes are mediated by the uneven distribution of productive assets across firms.

As referenced above, the identification strategy borrows from Atalay et al. (2019), who also rely on spatial variation to quantify the costs of crossing firm boundaries. Other common features include Poisson estimation and the use of merger evidence for validation. One key difference concerns scope. Atalay et al. (2019) analyze restricted Census data covering many industries and emphasize industry-level heterogeneity in costs. This paper focuses on a single industry in which transactions are relatively homogenous and emphasizes relationship-level variation. A second difference is that this paper aims to go beyond measurement by illustrating the importance of transaction costs for policy counterfactuals.

Methodologically, I adapt the standard logit matching model from the marriage literature (Choo and Siow, 2006) to model resource allocation within and across firms. This allows me to capture internal reuse, sharing, and subsurface disposal within a common framework in which substitution across the firm boundary arises naturally. Moreover, the model admits a highly tractable equilibrium characterization and (unlike popular bargaining models) is amenable to counterfactual analysis. A small number of papers in empirical industrial organization have previously adapted transferable utility matching models to study firm behavior: Fox and Bajari (2013) for spectrum reallocation, Akkus et al. (2016) for bank

mergers, and Fox (2018) for automobile supply chains. The models in these papers relax the restrictive additive separability assumption in Choo and Siow (2006) but require maximum score estimation. The model in this paper imposes additive separability in exchange for tractability: the main parameter estimates are obtained with Poisson pseudo-maximum likelihood estimation (Galichon and Salanié, 2024), while the counterfactuals are computed using stable and efficient algorithms (Galichon and Salanié, 2022).

Finally, this work directly contributes to the policy literature on the local environmental impacts of fracking. Hausman and Kellogg (2015) and Black et al. (2021) survey the environmental impacts of the so-called Shale Boom, including local impacts related to wastewater. Most relevant to this paper are studies that measure potential impacts from groundwater contamination (Muehlenbachs et al., 2015; Hill and Ma, 2022) as well as trucking-related externalities (Muehlenbachs et al., 2021). This paper highlights potentially significant challenges that can arise in designing policy to address the externalities identified in these papers. In this way, evidence of transaction costs is relevant for wastewater policy not only in Pennsylvania but in all regions where fracking occurs. The potential impacts of wastewater specifically and the applicable legal and regulatory frameworks are described in EPA (2016) and Groundwater Protection Council (2019).

2 Setting

This section provides additional institutional context and introduces the data.

2.1 Disposal and reuse

During fracking, water is blended with sand or other small particles and specialized chemicals before being injected into a well under pressure. In the period that I study, a typical Pennsylvania “completion” required more than one hundred thousand barrels of fracking fluid.² Once a well begins to produce hydrocarbons, a large share of this fluid returns to the surface along with groundwater from the target formation.³ Production is initially rapid but declines steeply as reservoir pressure drops. Thereafter, a typical well sustains low levels of hydrocarbon and wastewater production for many years. Most Pennsylvania unconventional wells (i.e., wells completed with fracking) target the Marcellus Shale formation. For a typical

²A “barrel” is a standard measure of liquid volume in the oil and gas sector. A barrel holds 42 gallons (159 liters). A single barrel is abbreviated “bbl” and 1,000 barrels “Mbbl.”

³Water produced during or immediately after completion is often referred to as “flowback,” while water produced later is more commonly known as “produced water.” I use the term wastewater to encompass both.

Marcellus well, lifetime wastewater production amounts to around 50% of fracking fluid volume (Kondash et al., 2018).⁴

Wastewater is highly saline and may contain organic compounds, metals, and naturally occurring radioactive materials from groundwater in addition to fracking chemicals. As a result, careful handling and specialized disposal are required by law (Groundwater Protection Council, 2019). In most regions of the United States, subsurface injection disposal is the primary method of wastewater disposal. Injection disposal involves the use of specialized wells to deposit wastewater deep underground beneath impermeable rock formations.

In Pennsylvania, rock formations suitable for injection disposal are scarce (McCurdy, 2011). This makes conventional disposal unusually costly compared to other regions. Figure 1 shows that the large majority of conventional disposal recorded in the data (described below) occurs at injection disposal wells located in eastern Ohio. The distance between Pennsylvania gas wells and Ohio injection wells can be significant, exceeding 300 miles in the far northeastern part of the state (shown in green). Most wastewater is transported by truck.⁵ Over such a distance, trucking costs can easily surpass \$5.00 per barrel.

Before wastewater can be reused it must be treated to an acceptable standard. This standard can differ depending on exactly how the fracking fluid is designed. However, treatments tend to be simple and inexpensive, such as basic filtering and pH adjustment. Total costs of \$0.50-\$1.00 per barrel are typical (Oklahoma Water Resources Board, 2017). By comparison, injection disposal fees are often around \$1.00 per barrel (Puder and Veil, 2006; Tavakkoli et al., 2020). Moreover, treated wastewater directly substitutes for freshwater that would otherwise be pumped and transported from nearby sources at some cost.⁶ Net of these savings, treatment costs are rarely large enough to offset the transportation cost savings from reuse when the next best alternative is out-of-state injection disposal.

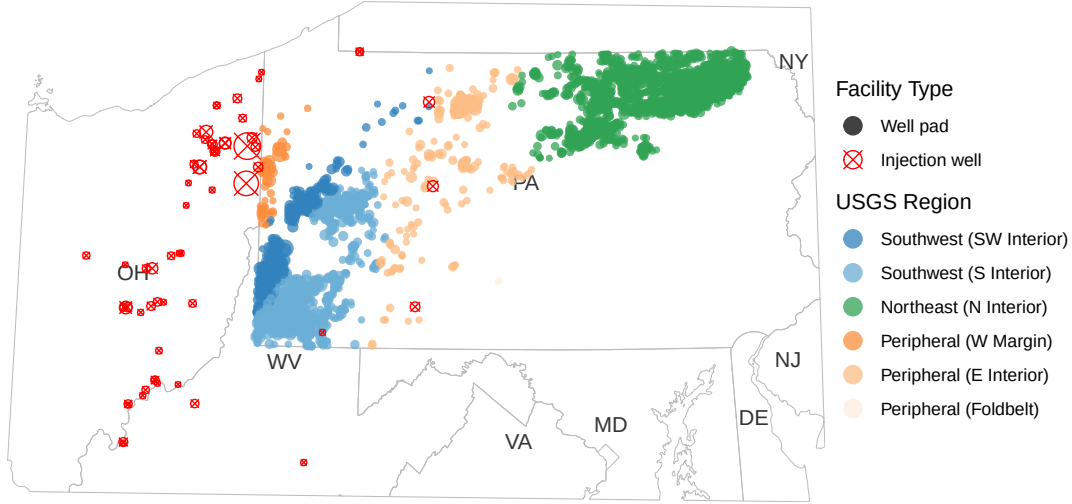
Firms often drill multiple wells from a single “well pad” in order to economize on infrastructure such as access roads and connections to gathering pipelines. Treatment for reuse can occur in one of two locations: either directly on a well pad where new wells are being completed, or at a dedicated facility known as a centralized treatment facility (CTF). While the former is slightly more common, both are used in Pennsylvania. CTFs facilitate storage and potentially allow for greater economies of scale in treatment. However, the use of CTFs tends to increase transportation costs as wastewater must be transported twice – once to

⁴In this respect the Marcellus region is quite different from other major shale plays in the United States such as the Permian in New Mexico and Texas. In “wet” shales, produced water volumes can easily exceed several multiples of fracking fluid volume (Kondash et al., 2018). Consequently, the maximum feasible rate of reuse outside Appalachia can be much lower than what is observed in Pennsylvania.

⁵The vast majority of wastewater is transported by truck, with alternatives such as pipeline and rail transportation accounting for less than 5% of produced water (Groundwater Protection Council, 2019).

⁶Pumping and transporting freshwater generally costs less than \$0.25 per barrel (Acharya et al., 2011).

Figure 1: Injection Well and Well Pad Locations



Notes: Locations of injection wells receiving wastewater from Pennsylvania unconventional wells, 2017-2020, in comparison to well pads reporting wastewater transfers. (1) For well pads, marker size corresponds to total wastewater disposition. For injection wells, marker size corresponds to total shipment volume. (2) Well pad marker colors: Author’s aggregation of USGS assessment unit from Higley (2019), assessment unit indicated in parentheses. See also Higley et al. (2019). (3) Gray lines indicate state borders.

the CTF, and then again to a well pad where it can be reused. There are also differences in regulatory treatment (such as bonding requirements).

From an environmental perspective, reuse usefully mitigates water-related externalities (Cooper et al., 2022). While often regarded as “safer” than alternatives such as surface water discharge or evaporation, injection disposal can cause seismic activity and create risks of groundwater contamination and surface spills (McCurdy, 2011; Groundwater Protection Council, 2019). Economically, subsurface pressure from injection disposal has the potential to raise local drilling costs (Bruant, 2023) and reduce productivity for nearby wells (McFarland, 2025). The Pennsylvania Department of Environmental Protection (DEP) “promotes [reuse] as a first option when it is technically and economically feasible” (PA DEP, 2024).

2.2 Pennsylvania wastewater records

The main data sources for this paper are well-level disposal records that oil and gas firms file each month with the DEP. These records indicate the quantity, disposal method, and destination of all waste materials transferred from each well site, including all wastewater.⁷ This

⁷Wastewater that does not leave the well site is not recorded. In particular, this means the data does not include wastewater that is reused on the same well pad where it is produced. This is potentially an

Table 1: Wastewater Disposition by Firm Type (Mbbl/Month)

	Southwest		Northeast		Peripheral	Total
	Small	Large	Small	Large		
Total Disposal	493.4	2449.4	477.8	1108.2	118.7	4647.4
<u>Destination Type</u>						
Well pad or CTF	306.5	2271.8	448.8	1077.8	15.5	4120.3
Injection well	133.9	86.5	29.0	22.9	103.0	375.2
Other	53.0	91.0	0.1	7.5	0.2	151.8
<u>Well Pad or CTF Type</u>						
Internal	47.3	2006.5	202.9	917.6	0.2	3174.5
Rival	100.5	65.8	153.8	54.8	12.7	387.6
3rd party	158.6	199.6	92.1	105.3	2.7	558.3
<u>Rival Type</u>						
Northeast (Large)	2.7	17.6	99.9	16.9	11.1	
Northeast (Small)	0.4	0.4	52.7	33.7	0.1	
Southwest (Large)	90.4	45.3	0.2	3.9	1.1	
Southwest (Small)	6.8	2.2	0.2	0.1	0.4	
Peripheral	-	0.1	-	0.0	-	

Notes: Author’s aggregation of Pennsylvania DEP waste report data for unconventional wells, 2017-2020. (1) Mbbl indicates one thousand oil barrels. (2) Classification of own and rival facilities based on identity of destination well pad operator at time of shipment. 3rd party CTFs facilities are centralized wastewater treatment facilities not operated by oil and gas operators. “Other” includes all shipments outside Pennsylvania (including shipments for reuse in West Virginia). (3) Southwest (Northeast) includes S Interior and SW Interior (N Interior) assessment units from Higley et al. (2019). All other regions are Peripheral. (4) Large firm defined as one of the five largest firms within a region based on gas production during the analysis period.

information provides for a rich but limited window into wastewater reuse: there are no prices and no details relating to intermediation such as the method of transport or contract terms. In practice, well-level waste attribution can be challenging when wastewater infrastructure (for example, a storage tank) is shared by wells on the same pad. I therefore aggregate the data to the well pad level. Beginning in 2017, transfers for reuse are clearly indicated along with identifying information for the destination facility. I hence analyze wastewater originating from unconventional wells in the period 2017-2020.

Table 1 summarizes wastewater disposition by destination type. Aggregating across all records, 88.7% of wastewater is transferred to a Pennsylvania well pad or CTF. In com-

important limitation of the data when multiple wells are drilled from the same pad or when wells are re-fracked, because it implies that transfer volume is potentially endogenous. The severity of the bias from ignoring this possibility depends on the time between fracking events and the cost of on-site storage.

parison, only 8.1% of wastewater is transferred to an injection well. A small amount of wastewater is transferred to other destinations, primarily well pads and CTFs in West Virginia and Ohio. Reuse is therefore the typical outcome, although some conventional disposal occurs. There are two major gas-producing regions within Pennsylvania, visible in Figure 1: one in the Southwest, and one in the Northeast. Due to the significant transportation costs associated with injection disposal, firms in the Northeast report much less injection disposal (3.3% vs. 7.5% for firms in the Southwest).⁸ Injection disposal is also less common for Large firms than for Small firms (3.1% vs. 11.9%).⁹

2.3 Wastewater sharing

Because wastewater production occurs gradually over the life of a well, the number of wells generating wastewater (a stock) is large in comparison to the number of active fracking operations (a flow). Consequently, reuse entails a many-to-one matching process. In the average month during the analysis period, 1,712.6 distinct well pads reported a wastewater transfer, but only 55.0 total completions occurred.¹⁰ The median well pad transferred only 0.4 Mbbl per month — less than 0.5% of the volume required to complete a new well. Conversely, a well pad appearing as destination in a particular month received an average of 47.4 Mbbl — about 50% of the requirement for a new well — from 31.4 distinct well pads.

For the purpose of the empirical model, I interpret the allocations that result from this many-to-one matching process as the outcomes of a one-to-one matching process in which individual truckloads of wastewater are matched to incremental units of water demand for specific fracking operations. This is a reasonable simplification provided that there are no strong complementarities across truckloads of wastewater.¹¹

Wastewater sharing can help to improve match quality in several circumstances. For instance, a small producer with no active fracking operations at a particular point in time might pay a local rival to take delivery of its wastewater. This type of transaction can significantly economize on transportation costs relative to Ohio injection disposal, while

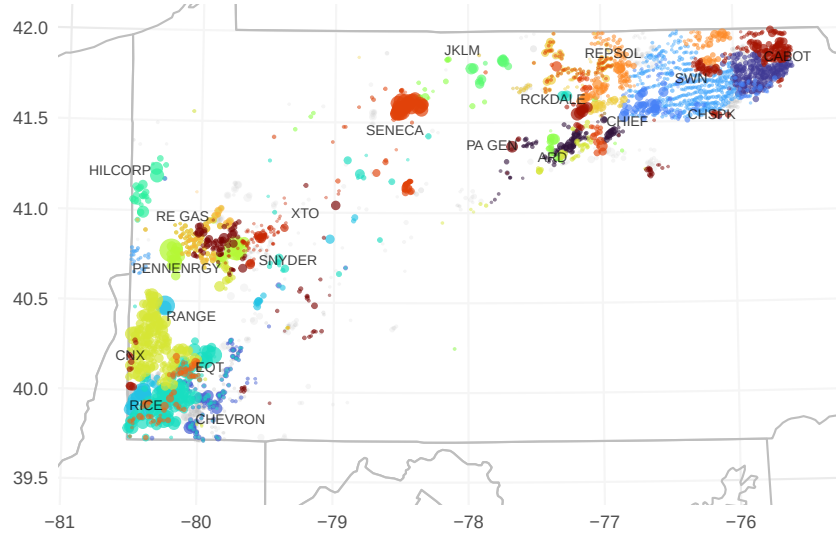
⁸Firms are assigned to either the “Southwest,” “Northeast,” or “Peripheral” regions as follows. First, I assign each well pad to one of the six Assessment Units from the USGS National Assessment Project for oil and gas resources (Higley et al., 2019). Next, I map Assessment Units to the Southwest, Northeast, and Peripheral regions as shown in Figure 1. Finally, I assign each firm to a region using the region in which the plurality of a firm’s well pads are located. In the table, note that injection disposal is the predominant method of disposal (86.8%) for Peripheral firms. This statistic is driven by a single firm.

⁹Large firms are defined as the five largest firms by gas production within a region.

¹⁰Calculation based on the average number of fracking jobs recorded in FracFocus during the sample period. By comparison, EIA’s Drilled but Uncompleted Wells (DUC) data implies 101.5 completions per month for the whole of Appalachia, including Ohio and West Virginia.

¹¹It may be violated if wastewater is differentiated and if there are scale economies in treatment. This possibility is formally excluded by the separability assumption in the model. See Section 4.

Figure 2: Well Pad Locations for Twenty Largest Firms



Notes: Locations of well pads reporting wastewater disposition from Pennsylvania unconventional wells, 2017-2020. (1) Limited to twenty largest operators by wastewater volume. Each color corresponds to a different operator. (2) Point size corresponds to total disposal volume.

avoiding the need for costly storage. Other producers might agree to share wastewater in a particular area in order to jointly minimize transportation costs. It is important to emphasize that the firms engaged in oil and gas extraction in Pennsylvania are highly heterogeneous: participants include the largest global energy producers (e.g., ExxonMobil, Shell, Chevron), large shale-focused firms (e.g., Range Resources, EQT, SWN), and numerous small firms, including small firms that may operate only a handful of wells (Small et al., 2014). Figure 2 plots the locations of the twenty largest firms, showing that individual firms tend to operate in specific geographic areas.

Table 1 distinguishes well pad and CTF destinations as “internal” or “rival” based on the identity of each origin and destination facility’s primary operating firm (inferred from DEP records). Some CTFs are operated by third party treatment firms rather than oil and gas producers.¹² These are designated as “third party” facilities in the table. Although the large majority of wastewater is transferred to internal or third party facilities, 8.3% of wastewater is transferred to well pads and CTFs operated by rival firms. This finding strongly suggests

¹²Producer-affiliated CTFs are often little more than semi-permanent systems of tanks or impoundments where the same treatments conducted on a well pad can be conducted at a larger scale. Third party CTFs are constructed similarly but may also be equipped with technologies that can treat water to higher standards, although these technologies are rarely used in practice.

that wastewater sharing plays an important role in facilitating wastewater reuse.

Unfortunately, the true extent of sharing activity is obscured by an important limitation of the data: for wastewater transferred to CTFs, the ultimate location of reuse is not recorded. Thus, wastewater might in principle be transferred from Firm A’s well pad to Firm B’s CTF for treatment before being re-transferred to Firm A’s well pad for use in fracking.¹³ To make progress despite this limitation, I assume re-transfers of this kind never occur. Similarly, I assume that wastewater initially transferred to internal or third party CTFs is never re-transferred to a rival firm.¹⁴ A “sharing transaction” is hereafter defined as a shipment to a rival well pad or CTF, granting that this definition entails some imprecision.

The middle columns of Table 1 disaggregate sharing activity defined in this manner by firm location and size. There are two key findings. First, sharing is more common in the Northeast (13.7% of wastewater) than the Southwest (6.4%). For firms in the Northeast, conventional disposal is more costly on account of the greater distance to Ohio injection wells. Hence, reuse in general creates more surplus, raising the probability that the benefits of sharing outweigh any transaction costs. Second, sharing is more common among Small firms (33.7% of wastewater) than Large firms (3.6%). Small firms complete new wells less often and therefore experience greater month-to-month variability in their need for water.

During the analysis period, 49 out of 75 operating firms shared wastewater on at least one occasion, including 9 of the 10 largest firms by wastewater production. On average, a firm that shared at least once shared in more than half of all months. Thus, most firms share and many do so often. Because it is costly to transport wastewater, distance appears to be an especially important factor in shaping matching patterns. For more than half of all wastewater, the recorded destination was among the five nearest facilities appearing as a destination for wastewater transfers from any origin in the same month. The average observed shipment distance was 30.0 miles, roughly 20% of the average distance between a randomly selected origin-destination pair. However, the distribution of shipment distances features a long right tail. At the 99%-tile, the shipment distance was more than 150 miles.

3 The EQT-Rice Merger

In November 2017 two of the largest shale gas producers in Pennsylvania merged, creating the largest single natural gas producer in the United States at the time. This section provides descriptive evidence that the merger substantially reduced the costs of coordinating

¹³The DEP prohibits “re-transfers” from well pads, so this concern applies only to CTFs.

¹⁴Without these assumptions, the “true” extent of sharing might vary from as little as 6% to as much as 42% of total disposal. The upper bound would require that 100% of transfers to a firm’s own CTFs are re-transferred to rival firms, which is highly implausible.

wastewater transfers for reuse, consistent with the elimination of transaction costs.

Prior to merging, EQT Corporation (“EQT”) and Rice Energy Inc. (“Rice”) both operated large numbers of wells on overlapping acreage in far southwestern Pennsylvania.¹⁵ In the six months leading up to the June 2017 merger announcement, 98% of Rice’s wastewater transfers originated from well pads within 20 miles of an EQT facility that appeared as a destination for wastewater transfers in the same month. 64% of EQT’s wastewater transfers originated at well pads within 20 miles of a Rice facility that appeared as a destination. Nevertheless, EQT and Rice did not share prior to the merger, even though EQT shared with firms other than Rice.¹⁶ This can be seen in Table 2, which summarizes wastewater transfers from former EQT and Rice well pads before and after the merger.

Table 2 also shows that transfers between formerly non-integrated EQT/Rice well pads increased sharply after the merger. Between November 2017 and the end of the sample, 62.4% of wastewater from former Rice well pads was transferred to former EQT well pads, and 22.5% of wastewater from former EQT well pads was transferred to former Rice well pads. Hence, the removal of the firm boundary was followed by substantial “sharing” across the pre-merger firm boundary. This outcome need not have occurred. For example, if the main factor preventing sharing before the merger had been technological rather than organizational (say, due to a mismatch between wastewater composition and fracking fluid design), transfers across the pre-merger firm boundary could have optimally remained zero.

Transaction costs are the shadow costs that explain why some economic activity takes place within firms rather than through markets (Coase, 1937). As organizing economic activity within firms is inherently costly, Coasean transaction costs are fundamentally net costs: the difference in organizational friction between arm’s-length and internal coordination. The substantive prediction made by the Coasean model is that transaction costs (understood as net costs) vanish once integration occurs. The changes in EQT and Rice shipments described in this section are consistent with this prediction, though they fall short of proving the existence of transaction costs. (For example, the merged firm might also have changed its fracking fluid design in ways that facilitated reuse.)

Many distinct economic mechanisms can generate transaction costs in the broad Coasean sense. In one classic taxonomy of transaction costs, Dahlman (1979) distinguishes between search and information costs, haggling and decision costs, and policing and enforcement costs. Costs falling into any of these categories could be present in the context of wastewater

¹⁵Locations of EQT and Rice facilities can be observed in the lower left corner of Figure 2. EQT was also present in West Virginia and northeastern Pennsylvania; Rice had a presence in Ohio.

¹⁶Rice did not send wastewater to any rival firms in the pre-merger period. Rice did receive a small amount of wastewater from well pads associated with Alpha Shale Resources. However, Rice and Alpha Shale Resources had formerly been partners in a joint venture.

Table 2: Wastewater Disposition for Former EQT/Rice Well Pads (Mbbl/Month)

	EQT Well Pads		Rice Well Pads	
	Pre	Post	Pre	Post
Total Disposal	521.4	516.6	245.1	292.7
<u>Destination Type</u>				
Well pad or CTF	520.9	501.2	245.0	283.3
Injection well	-	6.4	0.1	2.9
Other	0.5	8.9	0.0	6.6
<u>Well Pad or CTF Type</u>				
Pre-Merger EQT	434.4	187.6	-	95.1
Pre-Merger Rice	-	40.4	173.1	38.2
Post-Merger EQT/Rice	-	213.3	-	134.6
Other Rival	46.5	15.4	-	1.4
3rd party	40.0	44.5	71.9	14.0

Notes: Author’s aggregation of Pennsylvania DEP waste report data for unconventional wells, 2017-2020. For explanation of Destination Types, see notes to Table 1. (1) Mbbl indicates one thousand oil barrels. (2) “Pre” refers to the period preceding the merger announcement, defined here as 2017-01 to 2017-06. “Post” refers to the period following merger consummation, defined here as 2017-11 to 2020-12. (3) EQT Well Pads (alternatively Pre-Merger EQT) refers to well pads from which EQT reported wastewater disposition during the “Pre” period. Rice Well Pads (alternatively Pre-Merger Rice) refers to well pads from which Rice reported wastewater disposition during the “Pre” period. Post-Merger EQT/Rice refers to well pads from which EQT first reported wastewater disposition subsequent to the merger. Note that Rice ceased reporting wastewater disposition after the merger.

sharing. As explained below, the results of this paper provide only limited insight into the relative importance of different transaction cost mechanisms.

4 Model

This section presents the model. The model describes how firms minimize water-related costs, including the choice between internal reuse, sharing, and injection disposal.

Setup Let K_t denote the finite set of well pads endowed with wastewater in month t , and D_t the finite set of facilities endowed with capacity to accept wastewater. For now, D_t includes two types of facilities: well pads that need water for fracking, and injection wells. The key distinction is that firms operating well pads that need water for fracking potentially have preferences over the sources from which they receive wastewater (a two-sided market), while injection well operators do not (a one-sided market). CTFs are discussed below.

I assume that well pad $\kappa \in K_t$ is endowed with a continuum of truckloads of unstorable wastewater having mass $Q_{t\kappa}$. The capacity to accept wastewater at $\delta \in D_t$ is denoted by $C_{t\delta}$. When δ is a well pad, $C_{t\delta}$ is the maximum mass of wastewater that can be reused in ongoing fracking operations. When δ is an injection well, $C_{t\delta}$ is facility δ 's maximum monthly injection capacity. Each truckload of wastewater is matched one-to-one with an incremental unit of water demand at a well pad or an incremental unit of capacity at an injection well. For the remainder of this section I suppress the index t .

To describe the equilibrium, I denote the set of well pads that need water for fracking by D^1 and the set of injection wells by D^0 . I first consider the well pads that need water for fracking D^1 . The cost of using a truckload of wastewater i from well pad $\kappa \in K$ to satisfy the j th incremental unit of demand for water at $\delta \in D^1$ is given by

$$r_{\kappa\delta} + \phi_{\kappa\delta} - \epsilon_{i\delta} - \eta_{\kappa j}$$

where $r_{\kappa\delta}$ is the intrinsic cost of reuse, $\phi_{\kappa\delta}$ is a Coasean transaction cost, and $\epsilon_{i\delta}$ and $\eta_{\kappa j}$ are shipment-specific cost shocks incurred at the origin and destination facilities, respectively. The intrinsic cost of reuse $r_{\kappa\delta}$ includes all the costs that would be incurred if κ and δ were jointly managed by the same firm (such as transportation and treatment costs), while $\phi_{\kappa\delta}$ represents the net cost of crossing a firm boundary (i.e., the cost of sharing). By construction, $\phi_{\kappa\delta} = 0$ when facilities κ and δ are operated by the same firm.

In practice, some portion of the intrinsic cost of reuse is incurred by the sender while

some portion is incurred by the receiver: one or the other party must pay for trucking, for filters, for pH adjusters, and so forth. The incidence of costs is unobservable, but irrelevant for the equilibrium allocation under the assumption of transferable utility. I denote the sender's share of intrinsic costs by $r_{\kappa\delta}^K$ and the receiver's by $r_{\kappa\delta}^D$, where $r_{\kappa\delta} = r_{\kappa\delta}^K + r_{\kappa\delta}^D$. Similarly, some transaction costs $\phi_{\kappa\delta}^K$ may be borne by the sender and some by the receiver $\phi_{\kappa\delta}^D$, where $\phi_{\kappa\delta} = \phi_{\kappa\delta}^K + \phi_{\kappa\delta}^D$. Utility is transferable in the sense that private costs can be freely redistributed between the sender and receiver through transfer payments. I denote the (possibly negative) equilibrium transfer payment from the sender to receiver by $\tau_{\kappa\delta}$. Finally, I denote the total systematic cost of reuse by $\tilde{r}_{\kappa\delta} := \tilde{r}_{\kappa\delta}^K + \tilde{r}_{\kappa\delta}^D = r_{\kappa\delta} + \phi_{\kappa\delta}$ where $\tilde{r}_{\kappa\delta}^K := r_{\kappa\delta}^K + \phi_{\kappa\delta}^K$ and $\tilde{r}_{\kappa\delta}^D := r_{\kappa\delta}^D + \phi_{\kappa\delta}^D$.

Assumption 1. *If truckload i from $\kappa \in K$ fulfills the j th unit of demand for water at $\delta \in D^1$, the firm operating κ incurs a cost*

$$\tilde{r}_{\kappa\delta}^K + \tau_{\kappa\delta} - \epsilon_{i\delta} \quad (1)$$

and the firm operating δ incurs a cost

$$\tilde{r}_{\kappa\delta}^D - \tau_{\kappa\delta} - \eta_{\kappa j} \quad (2)$$

where $\tau_{\kappa\delta}$ is determined in equilibrium.

A characteristic feature of transferable utility matching models is that the sum of (1) and (2) does not depend on $\tau_{\kappa\delta}$. The crucial restriction imposed by Assumption 1 is additive separability of the cost shocks $\epsilon_{i\delta}$ and $\eta_{\kappa j}$.¹⁷ In particular, (i, j) -specific complementarities are excluded. Although transferring truckload i to δ may be more or less costly than transferring other truckloads from κ to δ , this difference must be the same for all units of capacity at δ . Similarly, the j th unit of demand at δ may be more or less costly to satisfy with wastewater from κ than other units, but this difference cannot vary across truckloads from κ .

For the firm operating well pad $\kappa \in K$, the alternative to reuse is injection disposal. Like transfers for reuse, transfers to injection disposal facilities can entail both intrinsic costs $r_{\kappa\delta}^K$ and transaction costs $\phi_{\kappa\delta}^K$. Unlike recipients of wastewater for reuse, however, injection well operators are unlikely to have preferences over truckloads of wastewater. I therefore assume that injection well operators accept shipments on a posted price basis without consideration for the characteristics of well pad κ (such as operator identity or wastewater quality).

¹⁷For an instructive discussion of separability see Chiappori (2017). In a recent contribution, Adda et al. (2025) propose a model which relaxes separability with marriage-specific Gaussian shocks.

Assumption 2. *If truckload i from $\kappa \in K$ is transferred to $\delta \in D^0$, the firm operating κ incurs a cost*

$$\tilde{r}_{\kappa\delta}^K + \lambda_\delta - \epsilon_{i\delta} \quad (3)$$

where λ_δ is determined in equilibrium.

Any sender that wishes to transport wastewater to δ incurs an endogenous common cost λ_δ . In equilibrium, this common cost rations the finite capacity for disposal C_δ at each injection well. It may encompass explicit disposal fees paid to the injection well operator and other economic costs that shift the cost of disposal at δ equally for all senders (e.g., unloading time), in addition to endogenous scarcity rents. Because almost all injection disposal facilities are owned by third party firms that do not engage in fracking, there is little observed substitution between internal and market-coordinated injection disposal. Hence, the data does not permit separate identification of $r_{\kappa\delta}^K$ and $\phi_{\kappa\delta}^K$ in the case of injection disposal. I therefore focus on the total systematic cost $\tilde{r}_{\kappa\delta}^K$.

For the firm operating well pad $\delta \in D^1$, the alternative to reusable wastewater is freshwater. The cost of sourcing one truckload of freshwater j for use at well pad δ is taken to be $\tilde{r}_{0\delta}^D - \eta_{0j}$. Freshwater sources are not observed in the data, so it is not possible to account for systematic heterogeneity in freshwater costs across the sources that can potentially supply well pad δ . If freshwater acquisition is subject to transaction costs in some circumstances, these transaction costs cannot be separately identified from intrinsic costs.

To close the model, I impose a logit structure on the idiosyncratic cost shocks.

Assumption 3. *For truckload i from $\kappa \in K$, the vector of cost shocks $\epsilon_{i\cdot} = (\epsilon_{i\delta})_{\delta \in D}$ consists of independent draws from an extreme value type I distribution with scale σ . For the incremental unit of water demand j at $\delta \in D^1$, the vector of cost shocks $\eta_{\cdot j} = (\eta_{\kappa j})_{\kappa \in \{0\} \cup K}$ consists of independent draws from an extreme value type I distribution with scale σ .*

This structure is found in Choo and Siow (2006)’s canonical model of the marriage market. As in one-sided discrete choice models, logit errors impose strong restrictions on substitution. Galichon and Salanié (2022) show that the equilibrium characterization presented below extends to a range of shock distributions that satisfy full support and finite expectation assumptions, including generalized extreme value distributions such as the nested logit. However, richer shock distributions come at a significant cost in terms of analytical and computational tractability. Moreover, Gualdani and Sinha (2023) show that sensible nonparametric restrictions on the shock distributions have limited identifying power for the underlying match surplus. I therefore proceed with the logit assumption.

Equilibrium The equilibrium concept is stable matching. A *feasible matching* assigns each truckload of wastewater from each well pad $\kappa \in K$ to some facility $\delta \in D$, while respecting all capacity constraints. A feasible matching is *stable* if there are no unmatched truckload-space pairs (i, j) such that matching would make both the sending and receiving firms (possibly one and the same) strictly better off given some transfer payment; no firm with a truckload i assigned for reuse would prefer to send i to injection disposal, and no firm with a space j assigned to accept wastewater would prefer to fill j with freshwater instead.

A matching can be described by a non-negative matrix $\mu \in \mathbb{R}_+^{|K| \times |D|}$ in which element $\mu_{\kappa\delta}$ represents the mass of shipments between κ and δ . The set of feasible match matrices $\mathcal{M}$ consists of all elements of $\mathbb{R}_+^{|K| \times |D|}$ that satisfy the following $|K| + |D|$ restrictions:

$$\sum_{\delta \in D^0} \mu_{\kappa\delta} + \sum_{\delta \in D^1} \mu_{\kappa\delta} = Q_\kappa \quad \forall \kappa \in K \quad (4)$$

$$\sum_{\kappa \in K} \mu_{\kappa\delta} \leq C_\delta \quad \forall \delta \in D^1 \quad (5)$$

$$\sum_{\kappa \in K} \mu_{\kappa\delta} \leq C_\delta \quad \forall \delta \in D^0 \quad (6)$$

Shipments of freshwater to $\delta \in D^1$ are denoted by $\mu_{0\delta} = C_\delta - \sum_{\kappa \in K} \mu_{\kappa\delta}$.

Under Assumptions 1-3, every stable matching minimizes expected private costs (inclusive of transaction costs) subject to the above feasibility constraints.¹⁸ There exists a unique matrix $\mu^* \in \mathcal{M}$ which describes all stable matchings (Gretsky et al., 1992; Galichon and Salanié, 2022). In particular, μ^* is the solution to a convex optimization problem:

$$\mu^* = \arg \min_{\mu \in \mathcal{M}} \langle \mu, \tilde{r} \rangle + \sigma \mathcal{E}(\mu) \quad (7)$$

The first term in (7) compactly denotes the sum of intrinsic and transaction costs incurred by senders and receivers across all transfers (reuse, injection disposal, and freshwater acquisition).¹⁹ $\mathcal{E}(\cdot)$ is a strictly convex entropy corresponding to the idiosyncratic cost shocks

¹⁸It is important to note that this is not a welfare result, because the welfare-relevance of transaction costs has not been established. See the discussion in Section 8.

¹⁹Formally, we have

$$\langle \mu, \tilde{r} \rangle = \sum_{\kappa\delta \in K \times D^1} \mu_{\kappa\delta} \tilde{r}_{\kappa\delta} + \sum_{\kappa\delta \in K \times D^0} \mu_{\kappa\delta} \tilde{r}_{\kappa\delta}^K + \sum_{\delta \in D^1} \mu_{0\delta} \tilde{r}_{0\delta}^D.$$

ϵ and η given μ .²⁰ Importantly, the solution μ^* does not depend on τ : transfer payments affect the division of surplus, but not the final allocation.

The structure of μ^* can be characterized as follows. First, fix an origin-destination pair $\kappa\delta \in K \times D^1$ representing potential transfers for reuse. In equilibrium, the firm managing κ 's demand to ship wastewater from κ to δ given the equilibrium transfer matrix τ is

$$Q_\kappa \exp \left\{ -\sigma^{-1} \left(\tilde{r}_{\kappa\delta}^K + \tau_{\kappa\delta} \right) - F_\kappa \right\} \quad (8)$$

while the firm managing δ 's demand to source wastewater from κ for reuse at δ is given by

$$C_\delta \exp \left\{ -\sigma^{-1} \left(\tilde{r}_{\kappa\delta}^D - \tau_{\kappa\delta} \right) - H_\delta \right\}. \quad (9)$$

These expressions follow from (1) and (2) together with Assumption 3, with F_κ and H_δ defined below. In equilibrium, (8) and (9) must exactly coincide. Equating (8) and (9) and eliminating $\tau_{\kappa\delta}$ gives $\mu_{\kappa\delta}^*$. Equilibrium injection disposal and freshwater transfers can likewise be derived from firms' facility-specific demand functions. Combining these results gives the following characterization of μ^* .

Proposition 4. *Given Assumptions 1-3, μ^* satisfies*

$$\mu_{\kappa\delta}^* = \begin{cases} Q_\kappa^{1/2} C_\delta^{1/2} \exp \left\{ -\frac{1}{2} \sigma^{-1} \tilde{r}_{\kappa\delta} - \frac{1}{2} F_\kappa - \frac{1}{2} H_\delta \right\} & \text{if } \kappa\delta \in K \times D^1 \\ Q_\kappa \exp \left\{ -\sigma^{-1} \left(\tilde{r}_{\kappa\delta}^K + \lambda_\delta \right) - F_\kappa \right\} & \text{if } \kappa\delta \in K \times D^0 \\ C_\delta \exp \left\{ -\sigma^{-1} \tilde{r}_{0\delta}^D - H_\delta \right\} & \text{if } \kappa\delta \in \{0\} \times D^1 \end{cases} \quad (10)$$

where (F, H, λ) are determined in equilibrium.

Equation (10) resembles a standard gravity equation (see, e.g., Head and Meyer, 2014). Extending the analogy, F_κ and H_δ correspond to the multilateral resistance terms in structural gravity models. In particular, F_κ is the logit inclusive value

$$F_\kappa = \log \left(\sum_{\delta \in D^0} \exp \left\{ -\sigma^{-1} \tilde{r}_{\kappa\delta}^K - \rho_\delta \right\} + \sum_{\delta \in D^1} \exp \left\{ -\sigma^{-1} \left(\tilde{r}_{\kappa\delta}^K + \tau_{\kappa\delta} \right) \right\} \right)$$

corresponding to (1), where it is convenient to define a normalized disposal cost $\rho_\delta := \sigma^{-1} \lambda_\delta$

²⁰Due to Assumption 3,

$$\mathcal{E}(\mu) = \sum_{\kappa \in K} \sum_{\delta \in D} \mu_{\kappa\delta} \log \left(\frac{\mu_{\kappa\delta}}{Q_\kappa} \right) + \sum_{\delta \in D^1} \left[\mu_{0\delta} \log \left(\frac{\mu_{0\delta}}{C_\delta} \right) + \sum_{\kappa \in K} \mu_{\kappa\delta} \log \left(\frac{\mu_{\kappa\delta}}{C_\delta} \right) \right]$$

for each $\delta \in D^0$. H_δ is the logit inclusive value

$$H_\delta = \log \left(\exp \{ -\sigma^{-1} \tilde{r}_{0\delta}^D \} + \sum_{\kappa \in K} \exp \{ -\sigma^{-1} (\tilde{r}_{\kappa\delta}^D - \tau_{\kappa\delta}) \} \right)$$

corresponding to (2). These objects, along with the normalized disposal cost vector $\rho = (\rho_\delta)_{\delta \in D^0}$, are pinned down by feasibility. This connection to gravity models motivates the estimation strategy described below.

Centralized treatment facilities (CTFs) When taking the model to data, I include producer-operated CTFs in the two-sided market via D^1 and third party CTFs in the one-sided market via D^0 . This treatment of producer-operated CTFs is reasonable if producers operate their own CTFs mainly in order to process wastewater for reuse at their own wells, such that substitution between alternative sources of water for fracking occurs before wastewater reaches the CTF. The treatment of third party CTFs is reasonable if third party CTFs accept wastewater non-selectively and if producers primarily view third party CTFs as a “black box” substitute for injection disposal. Like in the case of injection disposal, the transaction costs associated with the use of third party CTFs are not identified because third party CTF operators do not generate their own wastewater or frack new wells.

The main loss of realism from modeling CTFs in this manner is the omission of storage dynamics. This choice is forced by the data: I do not observe CTF re-transfers or inventory levels. The matching process is static within each month, and the facility-level demand for water is given exogenously. Supply-side dynamics are excluded: CTF re-transfers are not modeled, nor are drilling or fracking decisions that influence wastewater production.²¹

Interpretation and limitations The model crucially depends on strong price-taking assumptions. Firms make decisions on a truckload-by-truckload basis, both when sending and receiving wastewater. The firm can thus be viewed as a coalition of individual managers, one at each facility. Under price-taking, the equilibrium induced when managers at each facility myopically minimize costs coincides with the core (here, the stable matching). In the core, no coalition of managers can achieve lower costs than managers acting independently (Shapley and Shubik, 1971; Gretskey et al., 1992): hence, each firm minimizes total costs. In other words, the outcome coincides with the outcome that would be chosen if a price-taking firm centrally managed all of its facilities to minimize total cost. Transactions internal to the firm are “priced,” but these prices have a natural interpretation in terms of the shadow

²¹Some prior work on marriage markets has explored dynamic extensions of the basic transferable utility matching model (e.g., Choo, 2015).

price of scarce capacity.²²

The price-taking assumption conveniently integrates market transactions with the internal operations of the firm. However, it excludes firm-level strategic behavior such as capacity withholding. This type of behavior is often modeled using Horn and Wolinsky (1988)’s “Nash-in-Nash” bargaining model. Like the model in this paper, Nash-in-Nash presumes that firms jointly maximize surplus within relationships (here, conditional on transaction costs).²³ However, relationships may be formed strategically. One advantage of the matching framework is that no explicit model of relationship formation is required for counterfactuals.

Another significant limitation of the model is that all systematic costs are incurred on a per-truckload basis. In particular, there are no scale economies in transportation, treatment, or in transaction cost. Truck transportation costs are plausibly linear in volume. However, this assumption may be violated in the case of treatment costs (as evidenced by the use of CTFs). The absence of scale economies in transaction costs is arguably a more serious restriction. Many examples of transaction costs are best understood as fixed costs that might be incurred once for multiple shipments (e.g., search costs), or even once per relationship (e.g., the cost of drafting a first contract).²⁴ The estimates obtained from the model nevertheless remain informative as a reduced form when fixed transaction costs of this kind are present, since greater unit transaction costs are needed to rationalize the resulting distortions. However, unobserved fixed costs introduce bias in counterfactual simulations.

5 Estimation

This section describes the estimation procedure. I first derive a generic Poisson pseudo-maximum likelihood estimator via the method of moments. I then describe a two-step implementation of this estimator which limits the impact of measurement error.

5.1 Parameterization

Because freshwater acquisition costs in Pennsylvania are modest and less variable than injection disposal costs, I impose $\tilde{r}_{0\delta}^D = 0$ for all $\delta \in D^1$. The key primitives in the model are therefore the systematic costs of reuse and disposal and the dispersion parameter σ . For the purpose of estimation, it suffices to directly parameterize a set of normalized costs. In the

²²Reusing wastewater sourced from one well precludes the possibility of reusing wastewater from another, and the possibility of accepting transfer payments from other firms for use of the same capacity.

²³It is interesting to note that models of Nash-in-Nash bargaining typically do not account for transaction costs, either within relationships or in the process of relation formation (Carlton, 2020).

²⁴Parallels to this problem can be found in the trade literature when becoming an importer or exporter can entail fixed costs (e.g., Tintelnot, 2017).

two-sided market (on-pad reuse and producer-affiliated CTFs), I specify

$$\frac{1}{2}\sigma^{-1}r_{\kappa\delta} = \theta_d d_{\kappa\delta} + x'_{\kappa\delta}\theta_x \quad (11)$$

for the intrinsic costs of reuse and

$$\frac{1}{2}\sigma^{-1}\phi_{\kappa\delta} = w'_{\kappa\delta}\theta_w \quad (12)$$

for transaction costs.

In (11), $d_{\kappa\delta}$ is the distance between facilities κ and δ . The secondary cost vector $x_{\kappa\delta} = (x_{\kappa\delta}^{(1)}, \dots, x_{\kappa\delta}^{(I)})$ accounts for variation in treatment costs. In particular, it includes dummy variables which capture interactions between geological characteristics of the wells located at each facility (described below). The emphasis of geological characteristics is motivated by the concern that treatment costs may differ depending on wastewater composition and the optimal fracking fluid design, which largely depends on the target formation.

In (12), the covariate vector $w_{\kappa\delta} = (w_{\kappa\delta}^{(1)}, \dots, w_{\kappa\delta}^{(J)})$ accounts for the nature of the organizational relationship between κ and δ . In the simplest specification, $w_{\kappa\delta}$ is simply an indicator for the presence of a firm boundary (i.e., whether κ and δ are operated by rival firms). In richer specifications, transaction costs are allowed to vary across and possibly within relationships. As the cost of sharing could vary with wastewater composition or fracking fluid design, it may be informative to interact these features with geological characteristics.

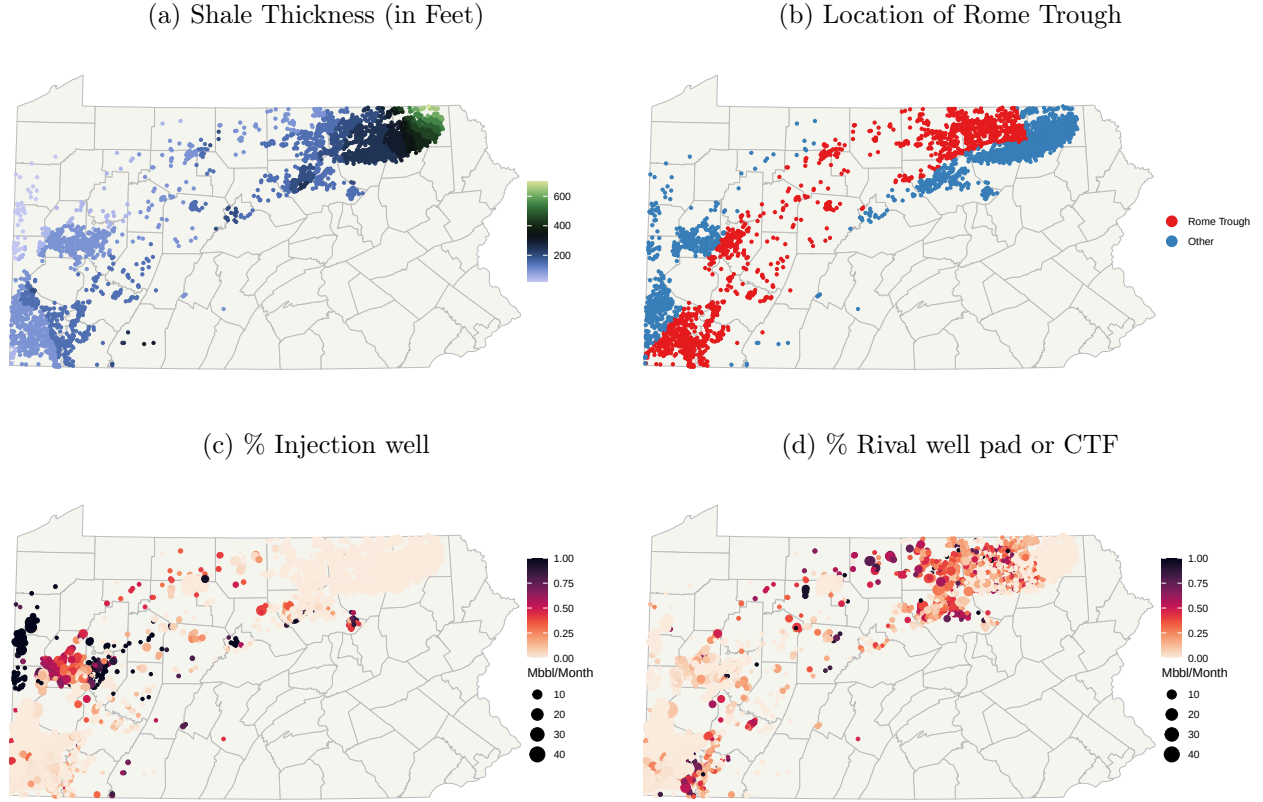
In the one-sided market (injection disposal and third party CTFs), I account only for distance. In particular, I specify

$$\sigma^{-1}\tilde{r}_{\kappa\delta}^K = 2\theta_d d_{\kappa\delta}. \quad (13)$$

The right hand side is multiplied by a factor of two so that the cost of distance is the same regardless of whether δ belongs to D^0 or D^1 .

Geological covariates I construct one set of covariates based on United States Geological Survey assessments of oil and gas development potential (Higley, 2019). These classifications are depicted in Figure 1. To capture additional variation, I include two covariates that may be especially important for wastewater composition and the optimal fracking fluid design: the local thickness of the Marcellus shale (Figure 3a) and the location of an important geological feature known as Rome Trough (Figure 3b). The former can directly influence the choice of proppant (sand) and chemical additives, while the latter predicts hydrocarbon characteristics via historical temperature and pressure exposures. Figures 3c and 3d depict

Figure 3: Key Geological Features of the Marcellus Shale, Wastewater Dispositions



Notes: Each circle represents a well pad appearing in the analysis sample. In (c) and (d), the size of each dot corresponds to total wastewater disposition. (a) Shale thickness from Energy Information Administration (2016a). (b) Rome Trough location from Energy Information Administration (2016b). (c) Percentage of wastewater disposition by well pad destined to injection well. (d) Percentage of wastewater disposition by well pad destined to rival well pad or CTF.

well pad-level disposal and sharing rates. Disposal is most common in thin-shale regions adjacent to the Ohio state line. Within the Northeast, sharing appears to be less common where the shale is particularly thick. Within both the Northeast and the Southwest, sharing appears to be more common above the Rome Trough.

Definition of “truckload” Shipment volumes are often reported in integer multiples of a truck capacities ranging between 80 and 130 barrels. The modal truck capacity among all such observations is 110 barrels. I therefore define a truckload as 110 barrels of wastewater.

Distance calculation I calculate over-the-road shipment distances using the Open Source Routing Machine (Luxen and Vetter, 2011) and data from OpenStreetMaps. I do not account

for roadway-specific vehicle weight or hazardous material restrictions that could alter optimal shipment routes for wastewater-hauling trucks in comparison to passenger vehicles.

5.2 Moment-matching procedure

I now reintroduce the time index t . K_t is defined as the set of well pads that report a positive volume of wastewater transfers in month t . D_t is defined as the set of facilities observed to receive a positive volume of wastewater transfers in month t , with injection wells and third party CTFs included in D_t^0 and well pads and producer-operated CTFs in D_t^1 .

Let $y_{t\kappa\delta}$ denote the observed quantity of wastewater shipments from κ to δ in month t , and let $\mu_t^\theta := \mu_t^\theta(Q, C)$ denote the equilibrium match matrix given parameters $\theta := (\theta_d, \theta_x, \theta_w)$ and capacities $Q_t := (Q_{t\kappa})_{\kappa \in K_t}$ and $C_t := (C_{t\delta})_{\delta \in D_t}$. For the moment, assume that Q_t and C_t are known. Under the parametric restrictions (11)-(13), Proposition 4 implies that

$$\log \mu_{t\kappa\delta}^\theta = \begin{cases} -\theta_d d_{t\kappa\delta} - x'_{t\kappa\delta} \theta_x - w'_{\kappa\delta} \theta_w + \frac{1}{2} \log Q_{t\kappa} - \frac{1}{2} F_{t\kappa} + \frac{1}{2} \log C_{t\delta} - \frac{1}{2} H_{t\delta} & \text{if } \kappa\delta \in K_t \times D_t^1 \\ -2\theta_d d_{t\kappa\delta} + \log Q_{t\kappa} - F_{t\kappa} - \rho_{t\delta} & \text{if } \kappa\delta \in K_t \times D_t^0 \\ \log C_{t\delta} - H_{t\delta} & \text{if } \kappa\delta \in \{0\} \times D_t^1 \end{cases} \quad (14)$$

where $\mu_{t\kappa\delta}^\theta$ is interpreted as the predicted mass of shipments from κ to δ in month t given parameter vector θ and marginal capacities Q_t and C_t .

To estimate the model, I impose that the difference between observed and predicted shipment mass is uncorrelated with observable facility-pair characteristics. For each $t = 1, \dots, T$, I impose the following moment restrictions:

$$E \left[(y_{t\kappa\delta} - \mu_{t\kappa\delta}^\theta) x_{t\kappa\delta}^{(i)} \right] = 0 \quad \forall i = 1, \dots, I \quad (15)$$

$$E \left[(y_{t\kappa\delta} - \mu_{t\kappa\delta}^\theta) w_{t\kappa\delta}^{(j)} \right] = 0 \quad \forall j = 1, \dots, J \quad (16)$$

$$E \left[(y_{t\kappa\delta} - \mu_{t\kappa\delta}^\theta) d_{t\kappa\delta} \right] = 0 \quad (17)$$

These assumptions may be violated if there is $\kappa\delta$ -specific unobserved heterogeneity in the systematic cost of reuse, distinct from the ij -specific unobserved heterogeneity excluded by Assumption 1. However, any κ - or δ - specific unobserved heterogeneity is addressed by the fixed effects $(F_t, H_t, \rho_t)_{t=1, \dots, T}$, where $F_t = (F_{t\kappa})_{\kappa \in K_t}$, $H_t = (H_{t\delta})_{\delta \in D_t^1}$ and $\rho_t = (\rho_{t\delta})_{\delta \in D_t^0}$.

I estimate the model by the method of moments. The sample moments corresponding to

(15)-(17) are given by

$$g_{xi}(\theta) = \sum_{t=1,\dots,T} \sum_{\kappa\delta \in K_t \times D_t^1} (y_{t\kappa\delta} - \mu_{t\kappa\delta}^\theta) x_{t\kappa\delta}^{(i)} \quad \forall i = 1, \dots, I \quad (18)$$

$$g_{wj}(\theta) = \sum_{t=1,\dots,T} \sum_{\kappa\delta \in K_t \times D_t^1} (y_{t\kappa\delta} - \mu_{t\kappa\delta}^\theta) w_{t\kappa\delta}^{(j)} \quad \forall j = 1, \dots, J \quad (19)$$

$$g_d(\theta) = \sum_{t=1,\dots,T} \sum_{\kappa\delta \in K_t \times D_t} (y_{t\kappa\delta} - \mu_{t\kappa\delta}^\theta) d_{t\kappa\delta}. \quad (20)$$

These moments pin down the $|I| + |J| + 1$ parameters in θ . Meanwhile, the sample analogues of the match feasibility restrictions (4)-(6) pin down the fixed effects $(F_t, H_t, \rho_t)_{t=1,\dots,T}$.

The use of the method of moments is motivated by the observation that the sample moment restrictions described above coincide with the first order conditions of a (weighted) Poisson pseudomaximum likelihood (PPML) estimator:

$$\begin{aligned} \max_{\theta, F, H, \rho} \quad & \sum_{t=1,\dots,T} \sum_{\kappa\delta \in K_t \times D_t^1} \{y_{t\kappa\delta} \log \mu_{t\kappa\delta}^\theta - \mu_{t\kappa\delta}^\theta\} + \sum_{t=1,\dots,T} \sum_{\kappa\delta \in K_t \times D_t^0} \frac{1}{2} \{y_{t\kappa\delta} \log \mu_{t\kappa\delta}^\theta - \mu_{t\kappa\delta}^\theta\} \\ & + \sum_{t=1,\dots,T} \sum_{\delta \in D_t^1} \frac{1}{2} \{y_{t0\delta} \log \mu_{t0\delta}^\theta - \mu_{t0\delta}^\theta\} \quad (21) \end{aligned}$$

The main estimates are obtained using a modified version of (21) which is described in the next section. At a high level, PPML is appealing because the wastewater data resembles typical trade datasets in which zero observed trade is frequent and errors are heteroskedastic. PPML is widely used in the literature on trade due to its favorable performance in such datasets and its relative simplicity (Santos Silva and Tenreyro, 2022). Moreover, PPML remains computationally feasible in the presence of high dimensional fixed effects.²⁵

5.2.1 Two-step estimation

Direct implementation of (21) requires knowledge of Q_t , C_t , and $(y_{t0\delta})_{\delta \in D_t^1}$. For the month t facility $\kappa \in K_t$ wastewater transfer volume, it is natural to specify

$$\hat{Q}_{t\kappa} = \sum_{\delta \in D_t} y_{t\kappa\delta}.$$

²⁵In particular, PPML is amenable to iteratively reweighted least squares (IRLS) estimation. In IRLS, the Frisch-Waugh-Lovell theorem is exploited to avoid inverting the full design matrix (Correia et al., 2020).

Hence, the mass of wastewater at κ in month t is represented empirically by the observed number of truckloads of wastewater transferred from κ in month t .²⁶

Specifying C_t is less straightforward. If $y_{t0\delta}$ were observed, one could specify

$$C_{t\delta} = y_{t0\delta} + \sum_{\kappa \in K_t} y_{t\kappa\delta}$$

for each $\delta \in D_t^1$. However, $y_{t0\delta}$ is not observed in the wastewater data. In the case that δ belongs to the two-sided market D_t^1 , $y_{t0\delta}$ can be estimated from mandatory fracking disclosures in the FracFocus database.²⁷ In Appendix A.1, I describe one possible estimation procedure based on this data. Yet any such procedure requires strong ad hoc assumptions and introduces sampling error, potentially contaminating the estimates of θ .

To mitigate this concern, I adopt a two step approach to estimation. Let $\tilde{C}_{t\delta} := \sum_{\kappa \in K_t} y_{t\kappa\delta}$ denote the sum of observed shipments to δ contained in the wastewater data. In the first step, I drop the freshwater margin from (21) and estimate

$$\max_{\theta, F, H, \rho} \sum_{t=1, \dots, T} \sum_{\kappa \delta \in K_t \times D_t^1} \{y_{t\kappa\delta} \log \tilde{\mu}_{t\kappa\delta}^\theta - \tilde{\mu}_{t\kappa\delta}^\theta\} + \sum_{t=1, \dots, T} \sum_{\kappa \delta \in K_t \times D_t^0} \frac{1}{2} \{y_{t\kappa\delta} \log \tilde{\mu}_{t\kappa\delta}^\theta - \tilde{\mu}_{t\kappa\delta}^\theta\} \quad (22)$$

where $\tilde{\mu}_{t\kappa\delta}^\theta := \mu_{t\kappa\delta}^\theta \left(\hat{Q}_t, \tilde{C}_t \right)$.²⁸ Given (14), C_t is absorbed by the destination fixed effects in (22). Hence, $\tilde{C}_t$ amounts to a convenient normalization which does not compromise the identification of θ .²⁹ (22) consistently estimates the main parameters of interest θ under the maintained assumptions. However, the fixed effect estimates $\left(\tilde{F}_t, \tilde{H}_t, \tilde{\rho}_t \right)_{t=1, \dots, T}$ from (22) do not estimate the population fixed effects $(F_t, H_t, \rho_t)_{t=1, \dots, T}$ from the infeasible model with $\mu_{t\kappa\delta}^\theta = \mu_{t\kappa\delta}^\theta \left(\hat{Q}_t, \hat{C}_t \right)$, for two reasons. First, swapping $\hat{C}_{t\delta}$ with $\tilde{C}_{t\delta}$ introduces a deterministic shift in the destination fixed effect for each $\delta \in D_t^1$. Second, the month t origin and destination fixed effects in (22) are only identified up to a scalar location parameter ζ_t .

In the second step, I use potentially noisy estimates of freshwater consumption to recover estimates of $(F_t, H_t, \rho_t)_{t=1, \dots, T}$ from $\left(\tilde{F}_t, \tilde{H}_t, \tilde{\rho}_t \right)_{t=1, \dots, T}$ given the population mapping

$$\left(\tilde{F}_t, \tilde{H}_t, \tilde{\rho}_t \right) = \left(F_t + \zeta_t, H_t - \zeta_t + \log \left\{ \frac{\tilde{C}_{t\delta}}{C_{t\delta}} \right\}, \rho_t - \zeta_t \right). \quad (23)$$

²⁶Per the discussion in footnote 7, wastewater reused on-site is not included.

²⁷FracFocus is a public registry in which firms disclose chemicals used in fracking particular wells (including water). In some states, disclosure is voluntary. In Pennsylvania, it is mandatory for unconventional wells.

²⁸To mitigate the impact of data reporting errors, I winsorize shipment volumes at the 99.9%-tile (about 77,000 barrels, or 600-700 truckloads in a month).

²⁹For the same reason, misspecification of $\hat{Q}_t$ due to unobserved on-site reuse does not bias $\hat{\theta}$.

As $\tilde{r}_{t0\delta}^D = 0$, freshwater demand at $\delta \in D_t^1$ as a function of ζ_t can be estimated by

$$\tilde{\mu}_{t0\delta}^\theta(\zeta_t) = \tilde{C}_\delta \exp \left\{ -\tilde{H}_{t\delta} - \zeta_t \right\}$$

Using this relationship, I estimate ζ_t by matching model-implied freshwater consumption to estimates $(\hat{y}_{t0\delta})_{\delta \in D_t^1}$ of freshwater consumption obtained from FracFocus disclosures:

$$\sum_{\delta \in D_t^1} \tilde{\mu}_{t0\delta}^\theta(\hat{\zeta}_t) = \sum_{\delta \in D_t^1} \hat{y}_{t0\delta}$$

Notably, the estimate $\hat{\zeta}_t$ depends on aggregate rather than facility-level freshwater consumption in month t , which may be more precisely estimated.

Given $\hat{\zeta}_t$, I then form $\hat{C}_{t\delta} = \tilde{\mu}_{t0\delta}^\theta(\hat{\zeta}_t) + \sum_{\kappa \in K_t} y_{t\kappa\delta}$ for each $\delta \in D_t^1$. This quantity is used for the counterfactuals. In the case of injection disposal wells or third party CTFs, I allow for slack capacity in counterfactuals by introducing fixed disposal costs that depend on the estimates $\hat{\rho}_{t\delta} := \tilde{\rho}_{t\delta} + \hat{\zeta}_t$ for $\delta \in D_t^0$. This strategy is explained in Section 7.1.

The two-step approach allows me to cleanly estimate the main parameters of interest θ separately from freshwater consumption. $(y_{t0\delta})_{\delta \in D^1}$ helps pin down the fixed effects, but this information is not necessary to learn about relative costs on the interior of the match.

Inference In the context of gravity models, Fernandez-Val and Weidner (2016) demonstrate that estimation of θ is not affected by the incidental parameters problem even when the number of fixed effects is large. However, the incidental parameters problem does contribute to biased standard errors and incorrectly centered confidence intervals for inference based on the score of the pseudolikelihood (Kauermann and Carroll, 2001). Zylkin (2024) proposes to address this issue using a two-step bootstrap. I adopt the same procedure.

5.3 Identification

Informally, there are three key objects to identify: the distance coefficient θ_d , the treatment cost parameters θ_x , and the transaction cost parameters θ_w . From (11) and (12), the systematic cost of a transfer from κ to δ can be expressed in the units of distance $d_{\kappa\delta}$ by

$$\frac{1}{2}\theta_d^{-1}\sigma^{-1}\tilde{r}_{\kappa\delta} = d_{\kappa\delta} + x'_{\kappa\delta}(\theta_d^{-1}\theta_x) + w'_{\kappa\delta}(\theta_d^{-1}\theta_w).$$

The distance-denominated treatment and transaction costs parameters $\theta_d^{-1}\theta_x$ and $\theta_d^{-1}\theta_w$ are identified by revealed preference-like arguments: these are differences in distance which equalize the systematic cost of reuse $\tilde{r}_{\kappa\delta}$ across potential transactions that are observably

different. Empirically, the key requirement is sufficiently large variation in observed shipment distance. The main transaction costs estimates below are presented in units of distance via $w'_{\kappa\delta}(\theta_d^{-1}\theta_w)$, which can be converted to dollars under specific trucking cost assumptions.

The distance coefficient itself is central to the interpretation of the estimated model. Fix $\theta_d^{-1}\theta_x$ and $\theta_d^{-1}\theta_w$ and recall that $\theta_d \propto \sigma^{-1}$, where σ is the scale of the idiosyncratic cost shocks ϵ and η . As $\sigma \rightarrow 0$, the equilibrium match matrix μ^* converges to the solution of a linear program in which the objective depends only on the systematic costs, as in (7) above. As $\sigma \rightarrow \infty$, the equilibrium match matrix μ^* converges to that of a random matching. Intuitively, θ_d is identified by the degree of randomness in the observed match.

Separation In some specifications that I consider, the coefficient vectors $x_{\kappa\delta}$ and $w_{\kappa\delta}$ can include many fixed effects. For example, I estimate models with firm-pair specific transaction costs. A general problem in PPML estimation (and other applications of maximum likelihood estimation) is that finite point estimates may fail to exist for certain parameters, particularly in the presence of high-dimensional fixed effects. For example, if there is no sharing within a given firm pair, the PPML estimate of the associated firm-pair specific transaction cost will diverge to $+\infty$. This problem is more generally known as the separation problem. With overlapping fixed effects, such as firm pair dummies combined with interactions between well characteristics, separation can manifest in subtle ways. To address it, I adopt an algorithm from Correia et al. (2021) to identify any separated “cells.” Because $y_{\kappa\delta} = 0$ for affected facility pairs and because $y_{\kappa\delta} \log \mu_{\kappa\delta}^\theta \rightarrow 0$ as $\mu_{\kappa\delta}^\theta \rightarrow 0$, I adopt the convention that $y_{\kappa\delta} \log \mu_{\kappa\delta}^\theta = 0$ and $\mu_{\kappa\delta}^\theta = 0$ when separation occurs. This approach is standard in statistical packages designed for gravity estimation (such as Correia et al., 2020).

Self-shipments 8.7% of wastewater disposal in the data originates at a well pad that is observed to receive wastewater in the same month: that is, for a pair of indices κ^* and δ^* corresponding to the same underlying facility, $\sum_{\delta \in D} y_{\kappa^*\delta} > 0$ and $\sum_{\kappa \in K} y_{\kappa\delta^*} > 0$. This is not surprising as the production of water is greatest immediately after a well is completed. Although it is never the case that $y_{\kappa^*\delta^*} > 0$ empirically, the model can predict $\mu_{\kappa^*\delta^*} > 0$.³⁰ The key issue is whether such shipments are physically feasible: wastewater from one well might be reused to frack another in the same location, but a well cannot be fracked with its own wastewater (at least, not in the course of the same fracking event). In the absence of more granular timing information, I constrain $r_{\kappa^*\delta^*} = \infty$ to exclude the second possibility. An unconstrained model would tend to significantly overstate the extent of on-site reuse.

³⁰Indeed, this will necessarily be the case for finite $r_{\kappa^*\delta^*}$ on account of the support of the logit errors.

Trucking costs Converting distance-denominated costs into dollars requires an estimate of the marginal cost of trucking. There is limited public information on the cost of wastewater hauling, which is typically intermediated at arm’s length between an operating firm and a wastewater trucking company. My preferred estimate is \$69 per hour, or \$2.50 per mile. This estimate is obtained by adding together estimates of driver wages and fuel costs and then adjusting for empty (“deadhead”) return trips (see Appendix A.2). The estimate is intended to be conservative: I do not account for vehicle depreciation, maintenance, overtime wages, or markups. Trucking cost estimates in the literature range as high as \$175/hour and \$4.3/mile (Puder and Veil, 2006; Tavakkoli et al., 2020).

6 Estimates

This section describes the model estimates. I first present preliminary estimates that establish the overall empirical significance of transaction costs. This exercise also provides the opportunity to explain in detail how the estimates are interpreted. I then investigate the extent of heterogeneity across transactions using a series of more flexible specifications.

6.1 Empirical significance of transaction costs

Table 3 presents parameter estimates for a series of parsimonious model specifications. Recall from Section 2 that the mean shipment distance is 30 miles. The estimates in column (4) imply that crossing a firm boundary increases the cost of reuse by an amount equivalent to the cost of shipping a truckload of wastewater $\frac{4.958}{0.036} = 137.7$ additional miles, a distance more than four times longer than the average shipment distance. This statistic can be viewed as a coarse measure of average transaction costs. If the cost of trucking is \$2.50 per mile (see Section 5), then crossing a firm boundary costs roughly \$3.50 per barrel of wastewater, substantially more than typical treatment costs (see Section 2).

To validate the interpretation of the estimates, the specification in column (4) also includes an indicator for post-merger transactions across the pre-merger firm boundary between EQT and Rice. The estimated coefficient is 0.184, consistent with a small positive cost of crossing the pre-merger firm boundary. Although statistically significant this cost is only 3.7% as large as the cost of crossing a firm boundary. I view this result as broadly consistent with a Coasean interpretation of the estimated firm boundary effects.³¹

Column (3) omits the firm boundary and EQT-Rice indicators from the model. This

³¹Residual within-firm friction is not surprising if post-merger integration is not completed immediately.

Table 3: Preliminary Model Estimates

	Dependent Variable: # of Truckloads					
	(1)	(2)	(3)	(4)	(5)	(6)
Distance	1.705* (0.017)	1.062* (0.017)	0.056* (0.001)	0.036* (0.001)	0.034* (0.001)	0.036* (0.001)
Firm Boundary		4.921* (0.072)		4.958* (0.064)		4.910* (0.059)
Firm Boundary x Active Relationship					2.342* (0.177)	
EQT-Rice Post-Merger		-0.017 (0.039)		0.184* (0.033)	0.205* (0.063)	0.186* (0.033)
Distance Measure	Log Miles	Log Miles	Miles	Miles	Miles	Miles
κ -month FEs	Yes	Yes	Yes	Yes	Yes	Yes
δ -month FEs	Yes	Yes	Yes	Yes	Yes	Yes
Geologic Controls	No	No	No	No	No	Yes
Deviance	5.86×10^6	3.22×10^6	5.76×10^6	2.97×10^6	2.45×10^6	2.92×10^6
R^2	0.500	0.753	0.486	0.763	0.787	0.765
Observations	7,607,893	7,607,893	7,607,893	7,607,893	3,292,034	7,607,893

Notes: Parameter estimates for the cost of reuse. Significance levels: $*p < 0.05$. (1) A “truckload” of wastewater is defined as a quantity of 110 barrels. (2) Distance refers to the distance between origin well pad and destination facility. (3) Firm Boundary is an indicator for whether the destination is a Well Pad or CTF not operated by the reporting operator at the origin Well Pad. Active Relationship is an indicator for whether any sharing occurs within a sender firm-receiver firm-month. (4) EQT-Rice Post-Merger is a dummy for transactions in the period 2017-11 and 2020-12 for which the origin is a Pre-Merger EQT well pad the destination a Pre-Merger Rice well pad or vice versa. See notes to Table 2 for further details. (5) κ -month and δ -month FEs refers to origin and destination fixed effects. (6) Geological controls include interactions between origin and destination facility characteristics. Characteristics include: (i) USGS assessment unit (Higley et al., 2019); (ii) indicator for overlay of Rome Trough; (iii) indicator for overlay of region with Marcellus shale thickness in excess of 350 feet. (7) R^2 calculated as squared correlation between predicted and observed dependent variable. (8) Standard errors obtained by 2-step bootstrap (Zylkin, 2024).

leads to a substantial drop in the quality of model fit, as measured by deviance and R^2 . The firm boundary is therefore not only economically large but quantitatively important for rationalizing the observed patterns of reuse. Moreover, comparing (3) and (4) shows that the estimated distance coefficient is substantially larger when the firm boundary is not taken into account. To help interpret this finding, columns (1) and (2) are estimated using log rather than linear distance.³² In this case the distance coefficient is interpreted as the negative of the elasticity of shipment volume with respect to distance. Dropping the firm boundary shifts the estimated elasticity from -1.062 to -1.705 . The first is much closer to typical distance elasticities estimated in the trade literature (including Atalay et al. (2019)’s study of trade between firms), which are often close to -1 .³³ The difference can be understood as follows. Firm boundaries are correlated with shipment distance. Without firm boundaries, the limited extent of sharing activity in the data is rationalized by very large distance elasticities; the implicit cost of distance is inflated relative to the scale σ of the idiosyncratic cost shocks. This mechanism plays a key role in the policy analysis below.

The firm boundary effect estimated from the specification in column (4) potentially combines two types of frictions: frictions in relationship formation (as in the case of EQT and Rice before their merger), and frictions within established sharing relationships. To assess the significance of the latter, the specification in column (5) interacts the firm boundary dummy with an indicator for whether the sending firm transferred a positive quantity of wastewater to the receiving firm in the same month, summing over all facilities. When this is the case, I say that firms have an active sharing relationship. Within active relationships, the firm boundary effect is equivalent to the cost of shipping a truckload of wastewater $\frac{2.342}{0.034} = 68.9$ additional miles, or about \$1.50 per barrel of wastewater. The estimated transaction costs thus remain economically and statistically significant on the intensive margin.³⁴

Column (6) augments the specification in column (4) with interactions between the geographical covariates introduced in Section 5. These terms are meant to address the possibility that apparent firm boundary effects are the consequence of intrinsic production costs which happen to coincide with firm boundaries (for example, due to wastewater composition or optimal fracking fluid design), rather than Coasean organizational frictions. If apparent

³²Comparing columns (2) and (4), linear distance improves model fit when accounting for firm boundaries. While log distance affords a convenient interpretation, linear distance may be more suitable in the wastewater management context if trucking contracts feature explicit linear elements (such as a cost per-mile or cost per-hour). The improvement in fit motivates me to use linear rather than log distance throughout the paper.

³³Head and Meyer (2014) report -1.1 to be the mean distance elasticity of trade across a large number of studies estimating structural gravity models. Chaney (2018) investigates this regularity. In their sample, Atalay et al. (2019) obtain an estimate of -0.96 .

³⁴Because firms that do not have an active relationship do not share, a large fraction of observations are separated by construction, reducing the effective sample size.

Table 4: Analysis of Variance for the Transaction Costs of Sharing

Specification	Deviance	R_1^2	R_2^2
1. Distance Only	3.51×10^6	0.000	
2. Distance + Rival Dummy	1.39×10^6	0.604	0.000
3. Distance + Time $\times$ Rival Dummy	1.38×10^6	0.607	0.007
4. Distance + Rival $\times$ Firm Dummies	1.26×10^6	0.641	0.093
5. Distance + Rival $\times$ Firm Dummies (Directed)	1.22×10^6	0.653	0.123
6. Distance + Rival $\times$ Firm-Pair Dummies	1.14×10^6	0.676	0.180
7. Distance + Rival $\times$ Firm-Pair Dummies (Directed)	1.09×10^6	0.689	0.214
9. Distance + Time $\times$ Rival $\times$ Firm-Pair Dummies	9.41×10^5	0.732	0.323
10. Distance + Time $\times$ Rival $\times$ Firm-Pair Dummies (Directed)	9.21×10^5	0.738	0.337
11. Distance + Rival $\times$ Firm Sizes (Interacted)	1.38×10^6	0.607	0.006
12. Distance + Rival $\times$ Firm Geologic Regions (Interacted)	1.37×10^6	0.610	0.014
13. Distance + Rival $\times$ Firm Sizes and Geologic Regions (Interacted)	1.36×10^6	0.612	0.019

Notes: Deviance (shipments for reuse only) under alternative transaction cost specifications. (1) All models include distance in miles. (2) R_1^2 refers to the McFadden pseudo R^2 with specification 1. as the baseline (i.e., one minus the ratio of the reported deviance and the deviance for specification 1.). R_2^2 defined analogously with specification 2. as the baseline. (3) Time refers to Month, the frequency of the wastewater disposition data. (4) Firm Dummy specifications include indicators for whether each operating firm is party to a transaction. Firm Dummy (Directed) specifications include separate indicators depending on whether a firm is the origin or destination in a transaction. (5) Firm-Pair Dummy specifications include indicators for each relationship. Firm-Pair Dummy (Directed) specifications include separate indicators depending on the direction of trade within the relationship. (6) Firm Size defined by quartiles of the firm size distribution, as measured by gas production during the analysis period. (7) Geologic Region refers to USGS assessment unit (Higley et al., 2019).

transaction costs are correlated with such costs, the standard error of the firm boundary coefficients should increase significantly in column (6). This does not occur, and model fit is not substantially improved.

6.2 Transaction cost heterogeneity

This section considers more flexible parameterizations of transaction costs. This allows me to examine how the effects of firm boundaries differ across relationships and over time. To this end, Table 4 reports model fit statistics from a sequence of models in which firm boundary effects are allowed to vary with transaction characteristics.

Relationship-level variation In the table, R_2^2 is a pseudo- R^2 statistic that indicates improvement in model fit relative to a baseline model with a single firm boundary dummy (i.e., a constant cost of crossing the firm boundary). Row 4 shows that interacting the firm boundary dummy with firm identity dummies generates considerable improvement versus

row 2 ($R_2^2 = 0.093$). This indicates that some firms experience higher transaction costs than others. Row 5 shows that there is further improvement when the effect of firm identity is allowed to vary between sending and receiving ($R_2^2 = 0.123$). Hence, the transaction costs a firm encounters when sending wastewater to rivals are generally different from those encountered on receipt. Row 7 shows the sending firm and receiving firm transaction costs from row 5 can differ substantially by counterparty ($R_2^2 = 0.214$). This implies that transaction costs are generally relationship-specific. Moreover, the difference between rows 6 and 7 indicates that transaction costs are frequently asymmetric within relationships depending on the sender ($R_2^2 = 0.180$ when transaction costs are assumed to be symmetric versus $R_2^2 = 0.214$ when they are not). These results illustrate that transaction costs exhibit substantial heterogeneity across relationships. In more formal terms, sharing appears to be mediated by an incomplete network of frictional bilateral relationships, where relationships are frictional in the sense that transaction costs can exist within them.³⁵

Correlation with relationship characteristics In principle, relationship-level variation in transaction costs can be useful for distinguishing between mechanisms that potentially contribute to transaction costs. Among available firm characteristics, however, I find little systematic correlation between relationship characteristics and transaction costs. Rows 11 to 13 of Table 4 report model fit statistics for specifications which allow firm boundaries to vary flexibility with firm location and size while omitting information related to firm identities.³⁶ Unlike the dummy variables in rows 4 through 10, neither firm size nor location substantially improves model fit ($R_2^2 < 0.02$).

Time-series variation The overall significance of transaction costs appears to vary little over time during the sample. Row 3 shows that interacting the firm boundary dummy with time dummies generates minimal improvement in model fit ($R_2^2 = 0.007$), suggesting that the initial firm boundary estimates from Section 6.1 are robustly estimated with respect to time. This finding also excludes a strong relationship between average transaction costs and market conditions, to the extent that these changed during the sample period.³⁷ On the other hand, rows 9 and 10 indicate that estimated transaction costs can change significantly within

³⁵This structure might be formalized in terms of a latent contract network with frictional trade (e.g., Fleiner et al., 2019). While such networks are rarely addressed in empirical industrial organization, they are ubiquitous in the real economy, motivating recent empirical work in international trade (e.g., Martin et al., 2023; Miyauchi, 2024) and development economics (e.g., Macchiavello and Morjaria, 2021).

³⁶For firm size, I use quartiles of the firm size distribution, as measured by gas production. For location, I use the USGS assessment unit containing the plurality of a firm’s wells.

³⁷From 2017 to 2020, the price of natural gas was low and essentially flat. I cannot rule out the possibility that the significance of firm boundaries might differ substantially under alternative market conditions.

relationships ($R_2^2 = 0.323$ versus $R_2^2 = 0.180$ in row 6 for the symmetric case, $R_2^2 = 0.337$ versus $R_2^2 = 0.214$ in row 7 for the asymmetric case).

7 Equilibrium impacts of disposal fees

In recent years, concerns over potential environmental and economic impacts from subsurface injection disposal have motivated regulators in key oil and gas producing states to restrict permitting and/or daily injection rates for injection wells, including in Oklahoma (Wertz, 2014; Money, 2019), New Mexico (Bennett, 2023), and Texas (Gough, 2025).

As discussed in Section 2, most injection disposal of wastewater originating in Pennsylvania is conducted in Ohio. In this section, I use the estimated model to examine the impacts of proposed amendments to an existing injection disposal tax in Ohio. This exercise demonstrates the importance of accounting for transaction costs in policy analysis.

Before presenting the results, I review the policy history and explain how the counterfactuals are constructed.

Policy history In 2010, the Ohio legislature instituted \$0.05 and \$0.20 per barrel fees on “in-district” and “out-of-district” disposal, respectively, with the latter covering all out-of-state disposal (mainly from Pennsylvania and West Virginia). Importantly, fees are assessed only on the first 500 Mbbl per well per year. Hence, high throughput disposal wells face a lower average fee. Ohio policymakers have subsequently sought to raise the disposal fee on two occasions. In 2012, Governor John Kasich unsuccessfully proposed to raise the out-of-district fee to \$1.00 per barrel. In 2023, the Ohio House of Representatives passed budget legislation that would have removed the 500 Mbbl threshold. This provision was eliminated after reconciliation with an Ohio Senate bill. Thus, the original fee structure remains in place despite continued interest in strengthening it. Meanwhile, Ohio residents have continued to express concern over the volume of injection disposal occurring in the state (Sour, 2025).

7.1 Counterfactual strategy

I analyze three policy counterfactuals: elimination of the existing fees, removal of the 500 Mbbl threshold, and a flat \$1 per barrel fee. There are two key challenges. First, I do not observe pass through of the existing out-of-district fee to Pennsylvania oil and gas firms. Second, I have not specified the capacity of existing injection disposal wells in Ohio, or how injection disposal capacity might change in response to changes in the tax rate.

To address the first problem, I simply assume that injection well operators pass through 100% of the expected average fee to their out-of-district disposal customers. This assumption is reasonable if the injection well sector is competitive and if marginal costs in the injection well sector are constant. Due to the 500 Mbbl threshold, average fees are not directly observable but can be estimated using well-level disposal records from the Ohio Department of Natural Resources. Appendix A.3 describes my preferred estimation strategy.

I address the capacity problem with the following indirect approach. To motivate the approach, consider that the expected cost of transferring wastewater from well pad $\kappa \in K_t$ to the lowest cost facility in the one-sided market is given by

$$V_{t\kappa}^{inj} = -\sigma c + \sigma \psi_{t\kappa}^{inj} \quad (24)$$

where c is Euler's constant and

$$\psi_{t\kappa}^{inj} := -\log \left(\sum_{\delta \in D_t^0} \exp \{ -\sigma^{-1} \tilde{r}_{\kappa\delta}^K - \rho_\delta \} \right) \quad (25)$$

is the inclusive value from the conditional cost minimization problem among facilities in D_t^0 . This follows from (3) and Assumption 3.

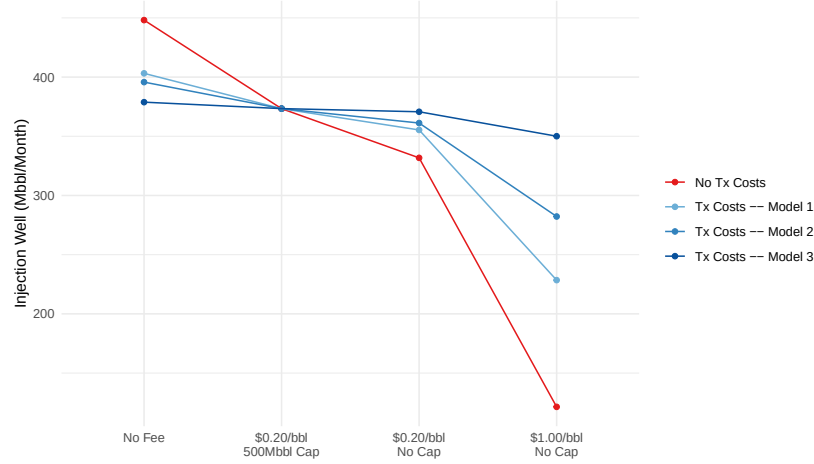
A useful observation is that the equilibrium of the two-sided market can be exactly reproduced by replacing the $|D_t^0|$ capacity-constrained outside options available to the firm managing κ with a single unconstrained outside option $\{0\}$ having (σ -normalized) cost equal to the inclusive value (25). For the counterfactuals, I directly adjust the empirical counterpart of (25) according to the counterfactual disposal fee schedule. In particular, I calculate

$$\hat{\psi}_{t\kappa}^{cf} = -\log \left(\sum_{\delta \in D_t^0} \exp \left\{ -2\hat{\theta}_d \left(d_{\kappa\delta} + \Delta_\delta^{cf} \right) - \hat{\rho}_\delta \right\} \right) \quad (26)$$

where Δ_δ^{cf} denotes a counterfactual change in the fee at $\delta \in D_t^0$ expressed as a distance premium.³⁸ I obtain counterfactual allocations by directly solving for the equilibrium of a two-sided market in which (26) is the normalized cost of the generic outside option $\{0\}$ at $\kappa \in K_t$. Counterfactual shipments to $\delta \in D_t^0$ can be recovered from facility-level logit probabilities conditional on the new equilibrium disposal volume $\hat{\mu}_{\kappa 0}^{cf}$.

³⁸I convert fees from dollars to distance using the trucking cost assumptions from Section 5. With the marginal cost of trucking equal to \$2.50 per mile, a \$0.20 per barrel fee has an equivalent cost impact on a Pennsylvania producer as moving all Ohio injection disposal wells 8.8 miles further away. A \$1.00 per barrel fee has an equivalent cost impact as moving all Ohio injection disposal wells 44.0 miles further away.

Figure 4: Injection Disposal Rates Under Counterfactual Ohio Injection Disposal Taxes



Notes: Injection well disposal volumes for selected model specifications in fitted model and Ohio disposal tax counterfactuals. (1) Model specifications: No Tx Costs refers to distance-only model (specification 1. in Table 4). Tx Costs – Model 3 refers to baseline model with time-varying asymmetric relationship-specific transaction costs (specification 10. in Table 4). Tx Costs – Model 1 refers to the constant transaction costs model (specification 2. in Table 4). Tx Costs – Model 2 described in the text. (2) \$0.20/bbl 500 Mbbl Cap refers to existing Ohio injection disposal well tax. No Fee is a counterfactual with no tax. \$0.20/bbl No Cap and \$1.00/bbl No Cap are counterfactuals without the existing fee schedule kink at 500 Mbbl.

This strategy allows me to predict the impact of counterfactual disposal fees in a localized sense without introducing a formal model of capacity investment in the injection well sector: I avoid directly specifying a capacity C_δ for any facility in D_t^0 , much as the availability of freshwater is left unspecified on the opposite side of the market. Capacity pressure on existing injection disposal wells with favorable locations is represented by the estimated endogenous costs $\hat{\rho}_\delta$ in (26), which remains fixed in the counterfactuals.

7.2 Predicted changes in disposal rates

Figure 4 plots the injection disposal rate (in Mbbl/month) under each disposal tax scenario. The dark blue line is obtained from a model that allows for time-varying, asymmetric, and relationship-specific transaction costs, while the red line is obtained from a model without transaction costs. The lighter blue lines capture intermediate cases discussed below.

Accounting for transaction costs substantially changes the predicted impact of modifications to the existing tax structure. With flexible transaction costs, the predicted reduction in disposal from the \$1.00 flat tax falls from 251.9 Mbbl/month (67.1% of status quo) to 23.3 Mbbl/month (6.2%). Likewise, the predicted impact of the existing tax structure falls from 74.8 Mbbl/month to 5.4 Mbbl/month. I attribute these findings to the bias in shipment

elasticities discussed in Section 6: transaction costs are positively correlated with distance, so failing to account for firm boundaries generates an upward bias in the magnitude of the shipment elasticity, equivalent in the model to a downward bias in σ . Thus, the model without transaction costs understates the extent to which the observed match is driven by idiosyncratic cost shocks, leading to overstated predictions of policy impacts.

A significant degree of flexibility is required to correct for this bias. Among the intermediate cases, Model 1 assumes a uniform firm boundary effect while Model 2 assumes infinite transaction costs for inactive relationships and zero transaction costs otherwise.³⁹ These might be viewed as reasonable but parsimonious specifications given the merger evidence from Section 3. Yet in each case, policy impacts are materially overstated relative to the more flexible model (dark blue in Figure 4), especially for the \$1.00 flat tax scenario.

8 Transaction costs and economic regulation

Oil and gas wastewater management potentially generates a variety of external costs, both environmental and economic. Most notable are perhaps the costs of increased seismic risk in the vicinity of injection wells. The use of injection wells for wastewater disposal (more so than fracking itself) has been linked to extensive low-level seismic activity in Oklahoma, Texas, New Mexico, Ohio, and other states. While past injection well-induced earthquakes have posed little danger to humans, the perception of seismic risk has the potential to significantly affect property values (Koster and van Ommeren, 2015; Gibbons et al., 2021). Economic regulation may be justified to ensure that injection disposal prices reflect these costs. Indeed, several of the state-level injection disposal restrictions referenced in Section 7 were prompted by well-publicized seismic events, including Ohio’s \$1.00 tax proposal.⁴⁰

Unpriced externalities are inherently inefficient, presenting a clear case for economic regulation. In contrast, the existence of transaction costs does not necessarily imply inefficiency. The key question for policymakers is whether transaction costs are welfare-relevant: that is, whether they constitute “true” costs or simply “decision” costs (see, e.g., Bernheim and Rangel, 2009). If transaction costs simply capture inherent costs of production (such as the wages of staff who might spend time coordinating wastewater shipments), there is little rationale for policy interventions to ease trade absent other market failures.

The empirical analysis in this paper does not distinguish between welfare-relevant and welfare-irrelevant transaction costs. Nevertheless, the estimated transaction costs are suf-

³⁹Hence, resources are assumed to be allocated efficiently among observed relationships. Assumptions of this kind are common when building equilibrium models of vertical markets in industrial organization.

⁴⁰A 2011 earthquake near Youngstown, OH was attributed to nearby injection disposal.

ficiently large to suggest that policies aimed at easing trade could significantly affect the size of other wastewater-related externalities. In a counterfactual with no transaction costs, injection disposal declines by 62% relative to the status quo. This magnitude suggests that policies that ease trade have the potential to lessen the need for conventional policy instruments such as corrective taxes, which may be difficult to implement optimally for various reasons. As in other contexts, a portfolio approach to policy can be useful when externalities interact with other market imperfections (Jaffe et al., 2005).

The remainder of this section considers specific policy instruments for reducing transaction costs before assessing potential rebound effects from making trade less costly. Given the clear environmental risks posed by injection disposal, I evaluate policies in terms of their equilibrium impacts on injection disposal specifically (setting aside other externalities).

8.1 Policy instruments

The estimates in Section 6 imply that there are two important types of transaction costs in the wastewater sharing market: those that prevent relationship formation, and those that raise the cost of trade within established relationships. To assess which is more important quantitatively, I construct a counterfactual in which transaction costs are eliminated only within active relationships (as defined in Section 6.1). In this case injection disposal declines by only 20%, substantially less than the 62% reduction from completely eliminating transaction costs. Thus, there appears to be greater scope for limiting injection disposal by easing relationship formation rather than reducing costs within relationships.

Two potentially important sources of friction in relationship formation are search and information costs on the one hand, and haggling and decision costs on the other (Dahlman, 1979). The first refers to the costs of identifying potential counterparties and evaluating gains from trade. Table 5 shows how the estimated transaction costs vary with the location of the sending and receiving firms.⁴¹ Transaction costs are elevated for cross-regional shipments, consistent with the presence of search and information costs. One way a state regulator can reduce search and information costs is by requiring complete and timely public disclosure of wastewater dispositions. Compared to Pennsylvania, other major oil and gas producing states such as Texas, Louisiana, New Mexico, and Oklahoma do not generally require detailed public disclosures of wastewater disposition. Another possibility is to require detailed reporting of all sources of base water in public completion reports. Based on my own analysis of Pennsylvania completion reports for the analysis period, source water

⁴¹First, I divided firms into five distinct groups: Southwest Large, Southwest Small, Northeast Large, Northeast Small, and Peripheral. Then I estimate a model allowing firm boundary effects to vary for each combination of sending and receiving firm type. Model fit statistics are reported in row 13 of Table 4.

Table 5: Estimated Firm Boundary Effects by Firm Region and Size

	To Southwest		To Northeast	
	Small	Large	Small	Large
From Northeast Large	182.8 (2.0)	157.6 (1.6)	107.3 (0.2)	163.0 (0.2)
From Northeast Small	192.6 (5.3)	221.6 (2.7)	122.5 (0.2)	136.5 (0.1)
From Southwest Large	157.2 (1.1)	150.1 (0.2)	208.6 (0.9)	127.9 (0.3)
From Southwest Small	134.4 (0.6)	131.2 (0.3)	207.3 (2.7)	192.6 (0.8)

Notes: Parameter estimates obtained by interacting firm boundary dummy with Firm Region and Size classifications. (2) Southwest (Northeast) includes S Interior and SW Interior (N Interior) assessment units from Higley et al. (2019). All other regions are Peripheral. (3) Large firm defined as one of the five largest firms within a region based on gas production during the analysis period. (4) Peripheral region omitted for concision due to limited sample size. (5) Standard errors obtained by 2-step bootstrap (Zylkin, 2024).

reporting is often non-specific (e.g., “Produced Water”) and sometimes incomplete.

Haggling and decision costs refer to the costs of writing contracts and bargaining over terms of trade. Groundwater Protection Council (2019) interviewed oil and gas firms operating throughout the country. One barrier to sharing highlighted in these interviews was legal and regulatory ambiguity regarding wastewater ownership, specifically on the question of whether wastewater is legally owned by the operating firm or the landowner. If wastewater is formally owned by landowners, “selling” wastewater may require consent of the landowner or royalty payments. This question currently remains unresolved in Pennsylvania.

A second issue highlighted in Groundwater Protection Council (2019) is legal and regulatory ambiguity surrounding liability for wastewater spills after wastewater crosses the firm boundary. If liability remains with the original operator, firms may be disincentivized from sharing, especially with riskier counterparties.⁴² To assess the plausibility of this mechanism, I calculate for each firm the rate of spill-related DEP enforcement actions per barrel of wastewater over the analysis period, focusing on enforcement actions that resulted in monetary penalties. I then divide firms into “more frequent violators” and “less frequent violators” based on the median rate among the ten largest firms. Finally, I re-estimate the model interacting sender and receiver types with a firm boundary dummy. The results are reported in Table 6. Consistent with potential liability concerns, transaction costs are larger when the receiver is a more frequent violator but differ little when the sender is a more frequent

⁴²Texas and Oklahoma have passed legislation to clarify liability in connection with reuse and disposal.

Table 6: Firm Boundary Effects by Frequency of Spill-Related Enforcement Actions

	To Less Freq. Violator	To More Freq. Violator
From Less Frequent Violator	123.3 (0.1)	162.5 (0.2)
From More Frequent Violator	124.4 (0.1)	176.2 (0.3)

Notes: Parameter estimates obtained by interacting firm boundary dummy with the frequency of spill-related DEP enforcement actions that resulted in monetary penalties during the analysis period. (1) For each firm, I calculate the rate of enforcement actions per barrel of wastewater. More Frequent Violators include a mix of large and small firms with violation rates exceeding the median rate among the ten largest firms. All other firms are Less Frequent Violators. (2) Standard errors obtained by 2-step bootstrap (Zylkin, 2024).

violator. This suggests that clarifying liability could incentivize greater sharing (and reduce injection disposal), though such a policy could have other less desirable effects as well.

8.2 Rebound effects

The analysis in this paper treats wastewater generation as an exogenous process. However, the volume of wastewater produced is ultimately a function of firms’ drilling and completion strategies. Policies like those described in the previous section tend to reduce the cost of disposal, potentially incentivizing increased drilling and hence increased wastewater generation (a type of “rebound” effect). In the extreme, such policies could even “backfire” in the sense of exacerbating the environmental harms they are designed to mitigate.

To investigate the magnitude of potential rebound effects, I use the fitted model to quantify producers’ cost savings from reuse and sharing under the status quo. Using a flexible specification for transaction costs, I predict that the average barrel of wastewater is shipped 93.1 miles fewer on account of reuse and sharing at current disposal costs.⁴³ This is reported in the final row of the Table 7. Under the assumption that the marginal cost of trucking is equal to \$2.50/mile, this implies shipping cost savings of \$2.12 per barrel of wastewater. On an output basis, savings of this magnitude amount to about \$0.020 per million British thermal unit (MMBtu) natural gas before including distance-invariant efficiencies from reuse. Such efficiencies are likely modest: perhaps \$0.50-\$1.00 per barrel of wastewater based on the figures cited in Section 2.1. Total cost savings (before transaction

⁴³To form this estimate, I compare the quantity-weighted average shipment distance in the estimated model with the quantity-weighted average shipment distance if reuse was not possible. To calculate the latter, I ignore injection well capacity constraints and hold fixed the estimated injection well costs.

Table 7: Quantification of Producer Surplus Gains from Reuse and Sharing

	Gas Production BCF/month	Wastewater Mbbl/month	Mean Shipment (mi)			Δ Surplus	
			No Reuse	Model	Δ	\$/bbl	\$/MMBtu
Northeast (Large)	181.3	1097.0	180.0	28.4	151.6	3.45	0.020
Northeast (Small)	53.1	475.2	180.2	39.0	141.2	3.21	0.028
Southwest (Large)	177.8	2333.3	94.3	25.6	68.7	1.56	0.020
Southwest (Small)	30.3	437.8	92.4	44.5	47.9	1.09	0.015
Peripheral	3.3	118.4	38.0	32.8	5.2	0.12	0.004
Total	445.9	4461.6	122.8	29.8	93.1	2.12	0.020

Notes: Counterfactual shipment distance and surplus estimates based on baseline model specification. (1) Gas production data from Pennsylvania DEP. (2) Southwest (Northeast) includes S Interior and SW Interior (N Interior) assessment units from Higley et al. (2019). All other regions are Peripheral. (3) Large firm defined as one of the five largest firms within a region based on gas production during the analysis period. (4) Surplus calculations assume the marginal cost of trucking is \$2.50 per mile. (5) BCF to MMBtu conversion assumes 1 Mcf $\approx$ 1.038 MMBtu.

costs) are therefore on the order of \$3.00 per barrel, or \$0.029 per MMBtu. This amounts to 1% the average spot price of natural gas at Henry Hub during the analysis period.⁴⁴

Newell et al. (2019) estimate a supply elasticity of 0.89 for US gas production (including conventional production). Hausman and Kellogg (2015) estimate a long-run supply elasticity of 0.81 for shale gas. These estimates imply that cost savings on the order of 1% of the benchmark price of gas are consistent with an increase in marketed production of less than 1%. This suggests that most observed production in Pennsylvania would have occurred (at higher cost) even if reuse and sharing had not been possible, with marginal wells being concentrated in the Northeast.⁴⁵ Even in the Northeast, Table 7 suggests that cost savings are no more than 2% of the benchmark price of gas. Given that injection disposal costs in the Northeast are extreme by industry standards, this suggests that future policymaking related to water and wastewater is unlikely to have a large direct effect on extraction.

9 Conclusion

The Shale Boom has significantly increased oil and gas wastewater volumes in the United States, raising questions about the environmental and economic impacts of conventional

⁴⁴Equal to \$2.69 per MMBtu based on monthly average spot price from EIA.

⁴⁵Put differently, past improvements in water efficiency due to reuse and sharing in Pennsylvania are unlikely to have “backfired” in the sense of causing more drilling to occur.

disposal practices like injection well disposal. Reuse is a promising alternative to conventional disposal, particularly when firms are able to share wastewater prior to reuse. However, the EQT-Rice merger evidence suggests that sharing can be impeded by Coasean transaction costs. I develop an empirical matching model to quantify the overall significance of these transaction costs in Pennsylvania, showing that they are empirically large, heterogeneous, and quantitatively relevant for wastewater policy. On the one hand, failing to account for transaction costs can lead to incorrect predictions of the effects of corrective taxes and other interventions. On the other, policies that aim to ease trade can be a useful tool for influencing environmental outcomes when direct Pigouvian regulations are difficult to implement.

The analysis exploits a rich and unusual dataset describing transactions both inside and outside the firm. This data permits the identification of Coasean transaction costs. Although I obtain highly granular estimates, the results provide limited insight into the specific economic mechanisms generating these costs. Without this insight, it is difficult to recommend specific policy interventions to address transaction costs, though I find some evidence that supports general measures enhancing transparency and regulatory certainty.

Pennsylvania has seen more extensive adoption of reuse than any other region. This is a consequence of two factors: relatively small volumes of wastewater, and a scarcity of injection disposal capacity. Yet recent years have seen a tightening of wastewater regulation in other regions such as the Permian Basin in Texas and New Mexico, where wastewater volumes are large and injection disposal capacity has historically been plentiful. These developments are likely to spur greater reuse. The results of this paper highlight the importance of accounting for transaction costs when evaluating wastewater policy, and more generally the importance of market structure and the distribution of control rights for environmental outcomes.

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A Appendix

A.1 Freshwater usage estimates

$\hat{y}_{0\delta}$ is constructed as the sum of observed wastewater shipments divided by an estimated firm-year level wastewater consumption rate. The latter is estimated by the difference between the sum of “TotalBaseFluid” for all fracking events appearing in the FracFocus data and observed disposals. A well’s freshwater usage is equal to one minus its wastewater usage. Freshwater usage is observable for a subset of wells in Northeastern Pennsylvania in data from the Susquehanna River Basin Commission (SRBC). A regression of the true wastewater usage rate in the SRBC data on the wastewater usage rate calculated from FracFocus “TotalBaseFluid” yields a coefficient 1.13 (SE 0.2) and an R^2 value of 0.49.

A.2 Trucking cost assumptions

My preferred estimate is based on the following assumptions:

1. Trucker wages are assumed to be \$22.62 per hour based on the BLS estimate of the average hourly wage of Pennsylvania truckers in 2018.

2. Average speed is assumed to be 27.8 miles per hour based on the ratio of duration and distance along calculated optimal routes for observed shipments.
3. Tanker-truck fuel efficiency is assumed to be 13.5 gallons of gasoline equivalent per 100 miles unladen and 15.4 gallons per 100 miles laden (assuming vehicle weights of 30,000 and 70,000 pounds, respectively) based on Bureau of Transportation Safety figures. Assuming 50% utilization this implies fuel efficiency is 6.9 miles per gallon of gasoline, equivalent to 7.9 miles per gallon of diesel. Unlike dry van trucking, backhauls for produced water are rare due to the necessity of cleaning the tank between loads.
4. Diesel price is assumed to be \$3.361 per gallon based on the 2018 average diesel retail price reported by EIA for PADD1B (East Coast - Mid-Atlantic).

A.3 Average disposal tax estimation

I obtain well-month level injection disposal volumes from the Ohio DNR. This data does not distinguish between in-district and out-of-district volumes. I therefore consider two possible approaches to calculating average fees at the well level. One possibility is to assume that 100% of observed disposal originates out-of-district. Another is to assume that only shipments observed in the Pennsylvania disposal data qualify as out-of-district disposals (thus assuming that the Ohio injection wells used by Pennsylvania firms receive no shipments from West Virginia or elsewhere).⁴⁶ Both measures produce similar (quantity-weighted) average fees: \$0.116 per barrel using the first approach, and \$0.135 per barrel using the second. Because the difference is modest, I adopt the first approach because it produces more conservative estimates of the impacts from the current disposal fee.

⁴⁶These measures coincide whenever I observe more than 500 Mbbl in annual shipments from Pennsylvania well pads to a particular injection well.